SECP2613-02-SYSTEM ANALYSIS & DESIGN

PROJECT REPORT

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ABSTRACT

The current project report showcases the concept and the creation of a modern, technologically integrated web application for the Kemubu Agricultural Development Board (KADA) Cooperative. The proposed system's objectives involve the automation of the existing manual methods of member registration and loan application, which are prone to errors. The following are potential problem areas: Data entry and data acquisition, data management, and human resource consumption. So it is necessary to improve the features and functions of the cooperative system as well as the design of a website to offer convenience to users to utilise KADA's cooperative benefits. Therefore, an analysis of current business workflow and core functionalities as a basic framework for designing a system to solve problems of current business flow by implementing advanced technology.

CHAPTER 1: INTRODUCTION

1.1 OVERVIEW

This chapter covers the initial steps in conducting this project. The proposed project aims to develop a technologically advanced web-based system for KADA's cooperative, streamlining member registration and loan applications to improve functionality and user experience. A background study was conducted to narrow down the topic so that it focuses on the important part. Next, a problem statement was identified after the main focus of the project was established from the background study so that a proposed solution to the problems could be made. After that, the objectives were set and the scope of the project was defined elaborating on the limitations of this study. Finally, based on all the components mentioned before, the benefits of the proposed system were determined before a summary was made.

1.2 BACKGROUND STUDY

The primary goal of this project is to propose a new, technologically advanced system that improves both the functionality of KADA's cooperative staff and the user experience for applicants. The new system aims to streamline the registration and loan application processes, reducing the manual workload and minimising errors, thus enhancing overall operational efficiency.

The proposed system design is informed by the principles and methodologies taught in the System Analysis & Design (WBS) course, which the team is currently undertaking in the second semester of study. Concepts such as Data Flow Diagrams (DFDs) provide a clear and structured overview of the new system. This theoretical foundation ensures the proposed solution is robust, efficient, and scalable.

This study is crucial for several reasons. Firstly, it provides the students with a practical application of their academic learning, allowing them to bridge the gap between theory and practice. Secondly, the implementation of a digital system for the KADA cooperative has the potential to significantly improve the agency's operations, benefiting both staff and farmers. By enhancing data accuracy, reducing processing times, and improving user accessibility, the new system can contribute to the overall development goals of KADA's cooperative.

1.3 PROBLEM STATEMENT

Inconsistency in Data Entry

KADA's data management issues are a significant concern, primarily due to inconsistent data formats and reliance on manual data entry. These inconsistencies can cause significant errors, as data entered manually is prone to human error. It is also inefficient to always fill out repetitive information such as employees' names and ID numbers for applications.

Data Management Issues

KADA uses paperwork as the main medium for membership and loan applications. Large volumes of documents can be hard to handle as they can be misplaced or lost during transfer or storage. Storage for the application forms also raises a problem as physical storage of documents requires space and that can lead to organisational challenges. Documents are also hard to retrieve when they need to be reviewed. It also raises security and privacy issues as sensitive information is involved

Limitations in Data Collection and Analysis

The manual method of collecting data will cause the data not to be categorised and not organised well. KADA expresses their difficulty in generating monthly and yearly reports of profits and losses. The lack of data analysis due to insufficient data can affect the corporation's decision-making process for prospects.

Reliance on Human Resources

The manual process heavily relies on large amounts of human resources as the whole membership and loan application process is extensive, making it labour-intensive and contributing to more costs. It also makes it more prone to human errors as various entities handle the entire process. There could be errors when reviewing applications to make sure the information provided is valid and failure to record payment details.

1.4 PROPOSED SOLUTION

The proposed system will be a website as an alternative to the manual system currently used by KADA. The system will still retain some functionality but with additional modifications that increase efficiency to enhance the working experience of the entities involved. The main improvement is digitising most of the processes of the current system such as filling in the membership and loan application. However, some system processes are intended to be kept the same, such as the board of directors meeting that KADA desires to be done online. An Information Systems expert as the admin will oversee the operation and maintain the system if needed. The website will have a database that users of the system can utilise to create, read, update, and delete data in the database. This addresses the data management issues where important data is stored in a safe place with a database. Only authorised personnel can have access to the data in the database.

Firstly, for existing employees of KADA who desire to apply for a membership application to be part of the cooperation. The system requires the employees to create an account before a membership application. Once the registration process is completed, the information will be stored in the database. When filling out the form, the system will have a function that controls data incompatible issues. For example, in Figure 2.1.1, when filling out the "YURAN DAN SUMBANGAN" section, the empty fields require an input of integers. If the user happens to input a value besides integers, the system will alert the user for a valid input instead. This is to combat inconsistent data entry issues. All payment processes will also be done through the website. A statement and receipt will be sent to the applicant as proof of successful payment. The applicants can also view their application status to see if they are accepted or rejected as cooperative members through the system.

As for the members of KADA, when applying for a loan, they only need to provide information not in the existing member database. Once they have provided relevant information, they can proceed with the payment process. The website also displays a chart of types of loans and their respective calculations according to the length of the loan application. The payment for loan application is a minimum of RM300 as a share modal and RM50 for the processing fee. If the loan application is rejected, the share modal will be refunded and the system will notify the applicants of the process. A statement and receipt will be generated that can be viewed by the user on the website. The applicants can also track their application status, the website will display the current process for loan approval. The system will provide the same feature to overcome the data management issue for the loan application.

From the viewpoint of the board of directors, they can access the applicant's information from the website. From there, they can approve the loan applications. Since the approval can be done within the website, this saves a lot of time because they can do it anywhere and anytime. The system will also provide monthly and annual reports on the number of new members and loan applicants. The reports will also provide details on profits and losses for the month and year. This information will be valuable to provide insights and analytics for when the board of directors holds a meeting.

Lastly, the admin will have the highest authority over our system. The admin has almost all the authority that members have. For example, the admin can update the salary and position of staff members and access personal information. The system also provides the admin to change the types of loans that can be applied. Other than that, records refund information, eases financial management compared to paperwork.

1.5 PROJECT OBJECTIVES

6.1 To analyse the user requirements of the system for KADA.
6.2 To design a member registration and loan application system for KADA.

1.6 PROJECT SCOPES

The study is limited to the digitalization of KADA's cooperative registration and loan application processes. The project aims to address the data management challenges faced by the KADA cooperative by designing and implementing a member registration and loan application system.

We are designing a comprehensive system for member registration and loan applications that replaces the current manual, paper-based process. In addition, provide automated data entry by implementing automated data entry features to reduce manual input, minimise errors, and improve efficiency.

Due to logistical constraints, all meetings with KADA representatives are conducted online. The initial meeting took place on 14th May 2024 via the Webex platform, with subsequent discussions held over WhatsApp to address ongoing questions and issues. This approach allows the team to gather necessary information and feedback efficiently, despite the lack of in-person interactions. The goal is to understand the background of the organisation and their needs. In addition, requirement gathering is conducted for thorough analysis of user requirements through KADA and observation to ensure the system meets the needs of KADA.

1.7 BENEFITS OF PROPOSED SYSTEM

The proposed system for the Kemubu Agricultural Development Board (KADA) is a comprehensive solution aimed at replacing the existing manual processes for membership registration and loan applications. This web-based platform ensures safe and effective data management by automating data entry, payment processing, and report generation. The online functionality of the system includes loan applications, data checking, automated notifications, report generation, and user registration. This solution will increase operational efficiency, lower mistake rates, and promote data management by automating these procedures. It lowers the possibility of data loss or misplacement by providing a centralised database for safe data storage and simple retrieval.

The proposed system improves accessibility by allowing KADA staff and members to interact with it from anywhere at any time with an internet connection. By combining different tasks, it makes the process easier for employees and users. The automatic creation of monthly and annual reports helps KADA management make better decisions by offering information for monitoring membership, loan applications, earnings, and losses. By reducing reliance on physical documentation and manual labour, the system significantly lowers operating costs. Powerful data protection mechanisms also tighten security and prevent privacy breaches. Easy navigation and broad acceptance are ensured by the user-friendly interface, which is in line with KADA's objectives to boost productivity, guarantee data accuracy, and assist the cooperative's goals for social and economic growth.

1.8 SUMMARY

The project aims to develop a technologically advanced system for the Kemubu Agricultural Development Board (KADA) to improve the registration and loan application processes. By digitising these processes, the system addresses issues such as inconsistent data entry, data management challenges, and the heavy reliance on human resources. Informed by principles from the System Analysis & Design (WBS) course, the solution includes features like automated data entry, secure data storage, and online payment processing. This will enhance operational efficiency, reduce errors, and provide easy access to data for staff and applicants. The system will also generate valuable monthly and annual reports, aiding decision-making for KADA's management. The project involves thorough user requirement analysis and online meetings with KADA representatives to ensure the system meets their needs, ultimately supporting KADA's goals for improved productivity, data accuracy, and social and economic development

CHAPTER 2: LITERATURE REVIEW

2.1 INTRODUCTION

This chapter further explains the purpose of this project. It provides a context as to how cooperative systems and information systems can be integrated to produce an efficient system. After that, a comparison between existing cooperative information systems from past research and studies was done to identify the methodology, technology, or systems used and how the digitalisation of the cooperative system contributes to the overall system.

2.2 INFORMATION SYSTEM

This section will discuss the definition of an information system and how cooperatives utilise information systems to boost their business efficiency.

2.2.1 DEFINITION OF INFORMATION SYSTEM

An information system is an integrated system of parts used to gather, store, process, and distribute data as well as knowledge and digital goods. Information systems are essential for businesses and other organisations to run and oversee their daily operations, communicate with suppliers and consumers, and fight in the market [3]. Electronic markets and inter-organizational supply networks are managed by information systems. Businesses employ information systems, for example, to handle their human resources, process financial accounts, and market to prospective consumers online.

2.2.2 COOPERATIVE INFORMATION SYSTEMS

Employee cooperatives are cooperatives formed and held by employees or workers of a firm or organisation. Employee cooperative's operational operations typically involve present or retired employees from the same company. Each member has the right to help manage the cooperative, including making choices about business activity and cooperative policies [5]. A Cooperative Management Information System extends the capabilities and effectiveness of a cooperative by implementing an information system that can fully automate the end-to-end tracking of cooperatives and their activities and guarantee the efficiency of all the changes in different modules [4].

2.3 PREVIOUS RESEARCH/SYSTEMS

Cooperative systems that will be compared are the Koperasi Pegawai Republik Indonesia (KPRI) and PT XYZ. Both (KPRI) and PT XYZ share a common goal of improving their business processes and information systems through structured architectural frameworks and recognising the necessity of a well-defined architecture to enhance operational efficiency and service delivery.

The two papers present techniques and different approaches to the architecture of the business & information system for two different organizations KPRI and PT XYZ. KPRI employs TOGAF (The Open Group Architecture Framework) in the Business & IT architectural strategy of the KPRI, especially in the preliminary phase of Vision architecture, requirement management, and establishing Data & Application architecture necessary to serve the Savings & Payment unit[2]. This method enables KPRI to guarantee an organisational and systematic route to assimilate and enhance their current Cooperative Information System (SIMASPRI) and website [2]. However, PT XYZ uses the Domain-Driven Design (DDD) approach in converting from a complex architecture to a microservices architecture for their loan application system [1]. This process includes the evaluation of business domains, identification of bounded contexts, and usage of the set of tactical DDD patterns that enable the construction of a system with fewer dependencies. In TOGAF's case, it underlies an extensive theoretical and practical foundation for enterprise architecture, however, DDD is more detailed and aims at developing complex software systems by implementing the microservices model, for instance, about eliminating dependency and increasing the pliability of systems when responding to various requirements and expectations [1]. Both approaches describe a plan for the improvement of their respective systems, but the methods as well as the focus of implementation are quite distinct.

2.4 SUMMARY

In essence, this chapter is done to improve the understanding of the difference between information systems and cooperative information systems. A comparison between past research contributes to how past implementation of information systems into existing cooperative systems improves the system's efficiency through methodologies used and advanced technology integrated into the system.

CHAPTER 3: METHODOLOGY

3.1 INTRODUCTION

This chapter covers the methodology used when executing the project. It also covers the phases involved when doing the project according to the methodology used. The project planning is also covered here gives details of the work that is done before starting with the project such as doing a feasibility study and allocating time before the due date of the project.

3.2 THE CHOSEN METHODOLOGY

The methodology used for this project is the waterfall method where each phase is done sequentially. Waterfall methodology requires a completed current phase before moving on to the next one. This method is used because it is suitable for smaller projects with deliverables that are easy to define from the start. This project only covers until the design phase of the system and the remaining phases will be continued in the following semester where the system will be implemented and tested.

3.3 PHASES OF CHOSEN METHODOLOGY

A. REQUIREMENTS PHASE

Due to logistical constraints, all meetings with KADA representatives are conducted online. The initial meeting took place on 14th May 2024 via the Webex platform, with subsequent discussions held over WhatsApp to address ongoing questions and issues. This approach allows the team to gather necessary information and feedback efficiently, despite the lack of in-person interactions. The goal is to understand the background of the organisation and their needs. In addition, requirement gathering is conducted for thorough analysis of user requirements through KADA and observation to ensure the system meets the needs of KADA.

B. ANALYSIS PHASE

From the interview, we analysed the components that are crucial for the system. From the information such as the scenarios or current business workflow, we made a Logical DFD AS-IS system (Context Diagram, Diagram 0) to determine the important components that will be implemented in the system. Analysing the system requirements such as functional and nonfunctional requirements are also done in this phase.

C. **DESIGN PHASE**

In this phase, we designed the proposed system, illustrated in the form of a Logical DFD AS-IS system (Context Diagram, Diagram 0, Child). The context diagram is further expanded into Diagram 0 and Child Diagram to further specify the process of the new system. Process specifications were done on the child diagrams of the processes to eliminate any redundancy or unclear processes in the child diagram. We also did partitioning, CRUD matrix analysis, event response tables for events that are present in the system, translating structure charts from the DFD, designing the system architecture and finally designing the expected interface of the system.

3.4 PROJECT PLANNING SCHEDULE

A project schedule is a timetable that arranges tasks, resources, and due dates in the most efficient order to ensure that a project is completed on time. A project schedule is prepared during the planning phase and comprises project timeline with start dates and end dates, the work necessary to complete the project deliverables the costs and the team members that are responsible for each task.

3.4.1 FINDINGS FROM FEASIBILITY STUDY

<u>Technical Feasibility</u>

The system will be a website that can run through various devices. The website requires devices, servers, and internet access. This is achievable as most people are well-equipped with the technology. After that, this website also needs a strong database system to safely store information. Database systems are also a well-known technology and well-versed among many people. Hence, an effective system with a good database system is also achievable. The information is crucial for the company to make insights and prospects to determine the next step for the benefit of the system.

<u>Operational Feasibility</u>

The website at any given time and anywhere since it can be accessed online provided an internet connection is available. The system will have features that reject incorrect data entries to reduce data errors improving system operations within the organisation. The system will also have a user interface that is easy to understand and operate to encourage users to use the system. Next, since the system is mostly automated, this will reduce the need for paper and labour workers, further improving the company's efficiency in operating.

3.4.2 COST-BENEFIT ANALYSIS

From the cost-benefit analysis (CBA) below, the Profitability Index is 1.17 which is more than one indicating that the project is economically feasible.

Assumption	
Sensasitivity factor (cost)	1.1
Sensasitivity factor (benefit)	1.1
Annual change in production cost	3%
Annual change in benefit	4%
Discount rate	10%
Estimated Cost	
Development cost	
Hardware	10000
Software development	10000
Documentation	1500
Production cost (per year)	
Maintenance	5000
Training	2000
Advertisement	1500
Transport	1500
IS Salaries	6000
Estimated Benefits	
Increase sales	24000
Savings	10000

Table 3.1 Assumption, estimated cost and estimated benefits for CBA

Costs	Year 0	Year 1	Year 2	Year 3
Development cost (one-time)				
• Hardware	11000			
• Software development	11000			
• Documentation	1650			
Total	23650			
Production cost (per year)				
• Maintenance		5500	5665	5834.95
• Training		2200	2266	2333.98
• Advertisement		1650	1699.5	1750.49
• Transport		1650	1699.5	1750.49
• IS Support		6600	6798	7001.94
Total		17600	18128	18671.8
Present Value (PV)		16000	14981.8	14028.4
Accumulated cost		39650	54631.8	68660.2

Benefits	Year 0	Year 1	Year 2	Year 3
• Increase sales		26400	27456	28554.2
• Savings		11000	11440	11897.6
Total		37400	38896	40451.8
Present Value (PV)		34000	32145.5	30392.1
Accumulated benefit		34000	66145.5	96537.5
Gain or Loss		-5650	11513.6	27877.3
Profitability Index (PI)	1.178743033			

Table 3.2 CBA Table

3.4.3 WORK-BASED STRUCTURE

A work breakdown structure (WBS) is a visual, hierarchical, and deliverable-focused breakdown of a project. It is a useful diagram for project managers since it allows them to divide their project scope and visualise all of the tasks needed to finish their projects. The table below shows the task allocation in our group. The WBS diagram can be referred to in **Appendix 8**.

Task / Name	Iman Abadi	Alif Fathi	Afif Shaqir	Afiq Danial
1.0 PLANNING				
1.1 Collecting information from an interview with representatives from KADA.	✔	✔	✔	✔
1.2 Determine scope of study.			✔	
1.3 Determining user requirements for the proposed system.	✔	✔	✔	✔
1.4 Meeting with team members to plan and divide tasks.	✔	✔	✔	✔
1.5 Choose the Methodology	✔	✔	✔	✔
1.6 Planning a schedule of the project	✔	✔	✔	✔
2.0 ANALYSIS				
2.1 Extracting key information from the interview.		✔	✔	
2.2 Determine problem statement based on interview.	✔			✔
2.3 Proposed solutions based on problem statement	✔	✔		
2.4 Determine objective of the study	✔	✔	✔	✔
2.5 Comparing the previous research.	✔			

2.6 Create a Project Planning Schedule	✔	✔	✔	✔
2.7 Summary the used method	✔			

Table 3.3 Tick table for Task Planning and Analysis

Task / Name	Iman Abadi	Alif Fathi	Afif Shaqir	Afiq Danial
3.0 DESIGN				
3.1 Describe the current business process (scenarios, workflow).	✔			✔
3.2 Determine functional and non-functional requirements of the system.	✔			
3.3 Design a logical DFD of AS-IS system.	✔	✔	✔	✔
3.4 Design a Logical DFD (TO BE) System	✔	✔	✔	✔
3.5 Design a Physical DFD (TO BE) System	✔	✔	✔	✔
3.6 Design a Partitioning				✔
3.7 Create a CRUD Matrix and System Architecture	✔			
3.8 Create an Event Response Table, and Structure Chart	✔	✔	✔	✔
3.9 Make a Interface Design / Prototype		✔	✔	

Table 3.4 Tick table for Task Design

3.4.4 PERT CHART

A programme evaluation review technique (PERT) chart is a graphical representation of a project's timetable that shows each job required to complete the project. The table includes the activities, predecessor and duration. The flow of the activity depends on its predecessor. If the predecessor is not yet completed, it cannot move on to the next activity. The duration just shows the amount of days taken to execute the activity. Refer to **Appendix 9** for the Pert Diagram

Activity		Predecessor	Duration (Days)
A	Collecting information from interview	None	3
B	Meeting with team member to plan and divide task	None	1
C	Extracting key information from the interview	A	1
D	Determine user requirement for the proposed system	B,C	2
E	Planning a schedule of the project	D	1
F	Choose the Methodology	E	1
G	Determine objective of the study	F	1
H	Comparing the previous research	G	1
I	Create a Project Planning Schedule	H,F	5
J	Determine problem statement based on interview	G	2
K	Proposed solutions based on problem statement	J	2
L	Describe the current business process	J	1
M	Determine functional and non-functional requirements of the system	K,L	1
N	Design a Logical DFD of AS-IS system	M	3
O	Design a Logical DFD (TO BE) system	N	3
P	Design a Physical DFD (TO BE) system	O	3
Q	Design a Partitioning	P	1
R	Create a CRUD Matrix and System Architecture	P	1
S	Create an Event Response Table, and Structure Chart	p	2
T	Make a Interface Design / Prototype	P,Q,R,S	5
U	Summary the used method	Q,T	2

Table 3.5 Table of activities for use in drawing a PERT diagram

3.4.5 GANTT CHART

A Gantt chart is a project management tool that allows managers to organise project schedules. It displays the order in which project tasks will be accomplished, as well as when they are due and how long they will take to complete. Every Gantt chart has two primary sections, a grid or task list on the left and a project timeline on the right.

The Gantt chart in **Appendix 5** shows our project's schedule and who is responsible for each task. It covers 32 days and breaks down all the tasks our team needs to complete. Each task has a start and end date, so we can see when we should be working on each part of the project.

Our team members, Afif, Afiq, Alif, and Iman, have different tasks to do, like collecting information, planning, and creating data flow diagrams. Some tasks are done together, and some are done individually. The chart helps us stay organized and make sure we finish everything on time.

3.4.6 GITHUB

This is the project repository of our project in GitHub. This serves as a way of communication between members and all the tasks are allocated here.

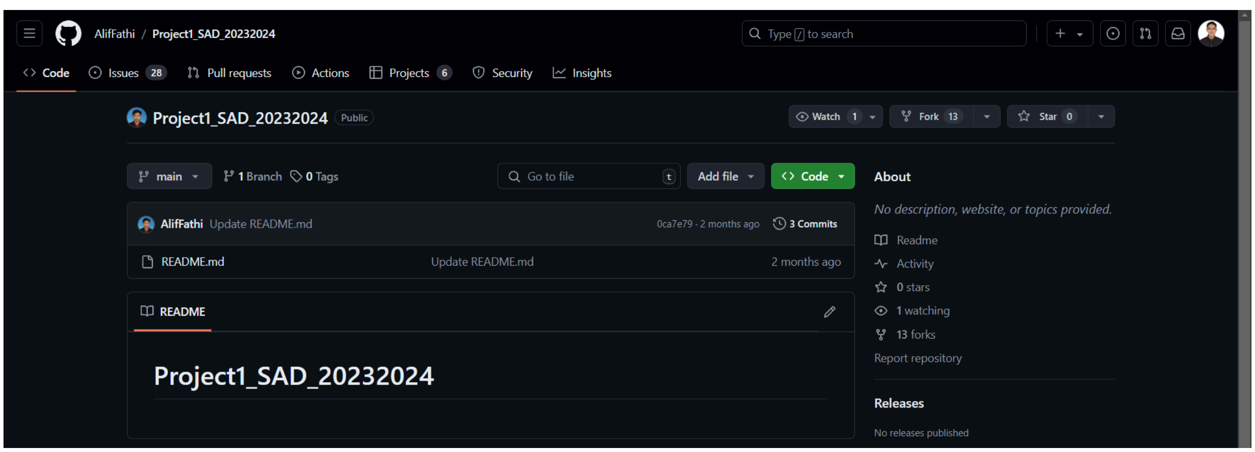


Figure 3.6 Project Repository

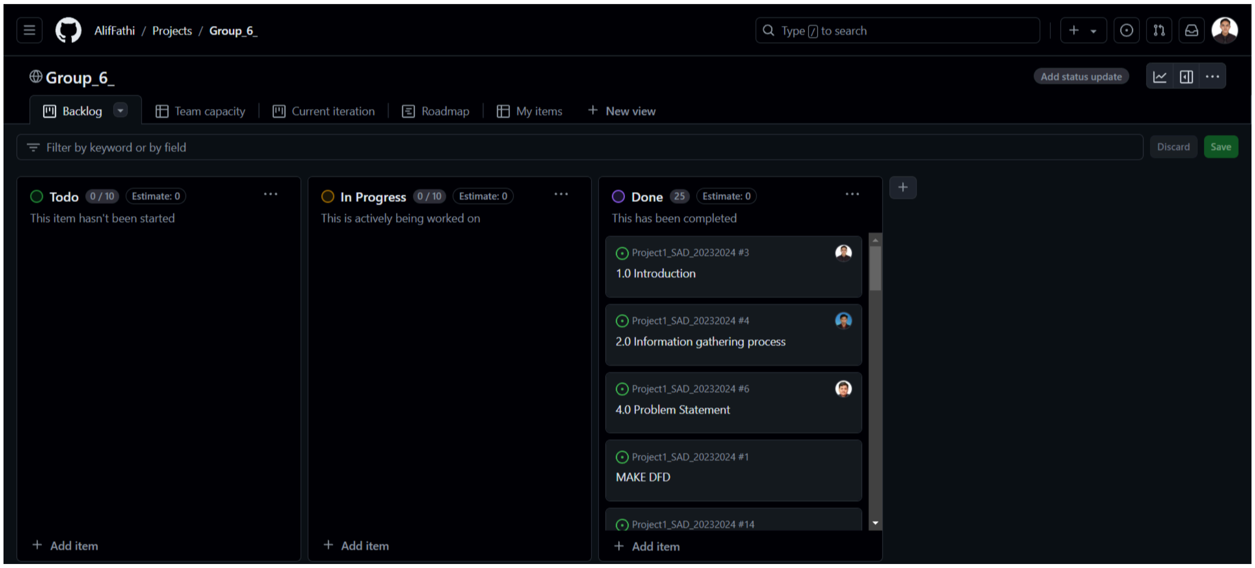


Figure 3.7 Backlog of the Activity

This is the backlog where all the planned tasks are recorded. We first list all the Todo tasks in its section. After that, the tasks are divided among us and we each need to show progress from the assigned tasks so that it is easy for other members to refer to the project flow. Then after the tasks are done, they will be listed in the Done section. The members will take up a new task until the project finishes.

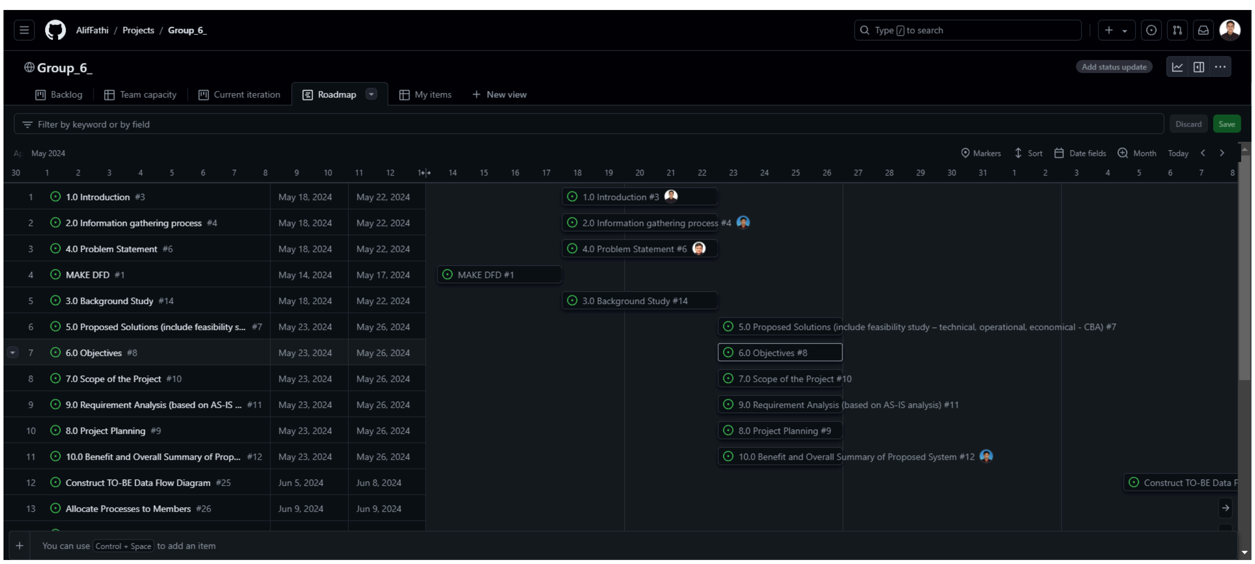


Figure 3.8 Project Roadmap

The project roadmap serves as a timeline for the project. Where all the tasks are listed while also showing the time taken for each task to be finished before moving on to the next one.

3.5 SUMMARY

In summary, this chapter discusses the methodology used to analyse the activities done in the phases according to the methodology. After that, a project planning schedule was done such as a feasibility study to determine the relevancy of the project. After that, the project planning part where time planning and task allocation were done to ensure the project ran smoothly. Task allocation was done to ensure that all members participated in the project.

CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

4.1 INTRODUCTION

This chapter covers the analysis and design part of the project. First, system requirements gathering was done to determine the system specification and business processes workflow of the current system. From there, the core components of the new system can be determined. A brief background of the company was provided to further aid in the system requirement-gathering process. From there, an illustration of the current business flow was done using digital flow diagrams to visualise and understand business and technical requirements. While still keeping the core components of the current system, a proposed system was built around it called the digital flow diagram (to-be) system detailing the new processes. A child diagram was used to explain the processes in a more detailed manner. Then, process specifications for a few ambiguous processes were done to obtain a precise description of the executed tasks. From there, partitioning was done to group processes together so that managing and processing large amounts of data becomes simpler. After that, A CRUD is used for capturing and displaying activities and permission within a system. The event response table was used for four different events to see how the system interacts with the entity. The structure chart was used to explain the child diagram and help visualise the system's complexity, making it easier to understand and manage. A diagram of system architecture was used to specify the technology used and how it is organised as it defines the system's boundaries, components, data flow, and communication channels of the system. Finally, the design of the system was translated into a non-working prototype using input and output design techniques to produce an inline user interface that meets up with the requirements of user experience goals.

4.2 SYSTEM REQUIREMENTS GATHERING

4.2.1 METHOD USED

For information gathering, first, we use an interactive method which is an interview. With an interview, we can seek information about KADA's history and how KADA manages their system. For the interview, we provide two types of questions which are open-ended questions and closed questions. An example of an open-ended question is “What is your opinion on your current system?” An example of a closed question is “How does KADA calculate their fee to apply for a loan?”

We also use an unobtrusive method which is to analyse quantitative documents. We analyse the payment records and forms to apply for membership and apply for a loan. With this data, we can find a variety of information such as the steps to apply for membership and how the payment system works.

By combining these methods, we aim to build a comprehensive understanding of KADA's historical context and current operational mechanisms. This will enable us to provide informed recommendations and improvements for the system.

4.2.2 SUMMARY OF FROM METHOD USED

Example Question :

1. Setakat ini, bolehkah saya menjejaki status permohonan pinjaman saya dalam talian?

2. Adakah terdapat jumlah maksimum pinjaman yang boleh kita pinjam?

3. Jika peminjam gagal membayar balik dalam tempoh tersebut, apakah penalti yang akan dikenakan?

4. Adakah terdapat kes di mana seseorang cuba memalsukan maklumat mereka, contohnya gaji mereka? Jika ada apakah tindakan yang diambil oleh pihak Koperasi KADA?

Other than interview questions, forms for loan application, membership application, financial statements of cooperative members and loan repayment chart were also provided by a representative of KADA's cooperative to further assist in gathering the system's requirements.

Refer from **Appendix 1 to Appendix 4** for examples of forms currently used for the current system.

4.2.3 CURRENT BUSINESS PROCESS (SCENARIOS, WORKFLOW)

A business process is a collection of actions or activities carried out by a group of stakeholders to achieve a common purpose. People or systems follow a structured process to achieve an established goal. Refer to **Appendix 6 and 7** for the flowchart showcases how the current business flow is done, providing us with the core component to design our system.

The first business flow is the membership application process where an employee needs to fill in a form for a membership application providing all the details required. After that, the application will undergo an approval process to determine their credibility. If they manage to become a member, they will have to pay a membership for the given time and the monthly afterwards.

When they have become a member, they have the privilege to apply for loans KADA cooperative offers. When applying for the loan they would need to apply using the loan application form providing all the details necessary for it. After that, they would need to pay a minimum of RM 300 for the share fee and a processing fee of RM50. After that, they will need to apply the approval process. If they are successful in applying for the loan, they will have to pay a monthly fee of a total from the loan repayment fee and the membership fee. If they are rejected, the share fee of RM 300 will be refunded to them.

4.2.4 COMPANY (COOPERATIVE OF KADA)

Kemubu Agricultural Development Board (KADA) is a pivotal government agency in Malaysia, tasked with fostering economic and social development among farmers in the Kemubu region. KADA has an association called KADA Staff Cooperative. A cooperative is an enterprise or organisation owned by and operated for the benefit of those using its services where worker cooperatives participate in the profits, oversight, and often management of the enterprise using democratic practices.

4.2.5 LOGICAL DFD AS-IS SYSTEM (CONTEXT DIAGRAM, DIAGRAM 0)

A logical DFD focuses on the business and how the business operates. It describes the business events that take place and the data required and produced by each event. Figures 4.1 and 4.2 show the data flow diagram designed on the requirements gathered.

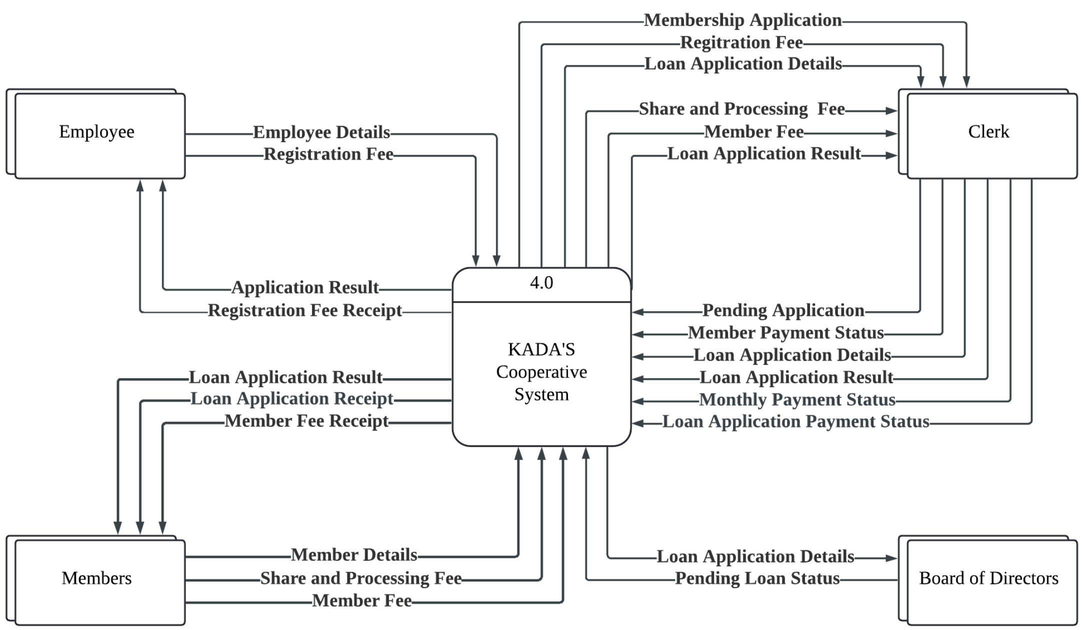


Figure 4.1 Context Diagram of KADA’s current Cooperative System

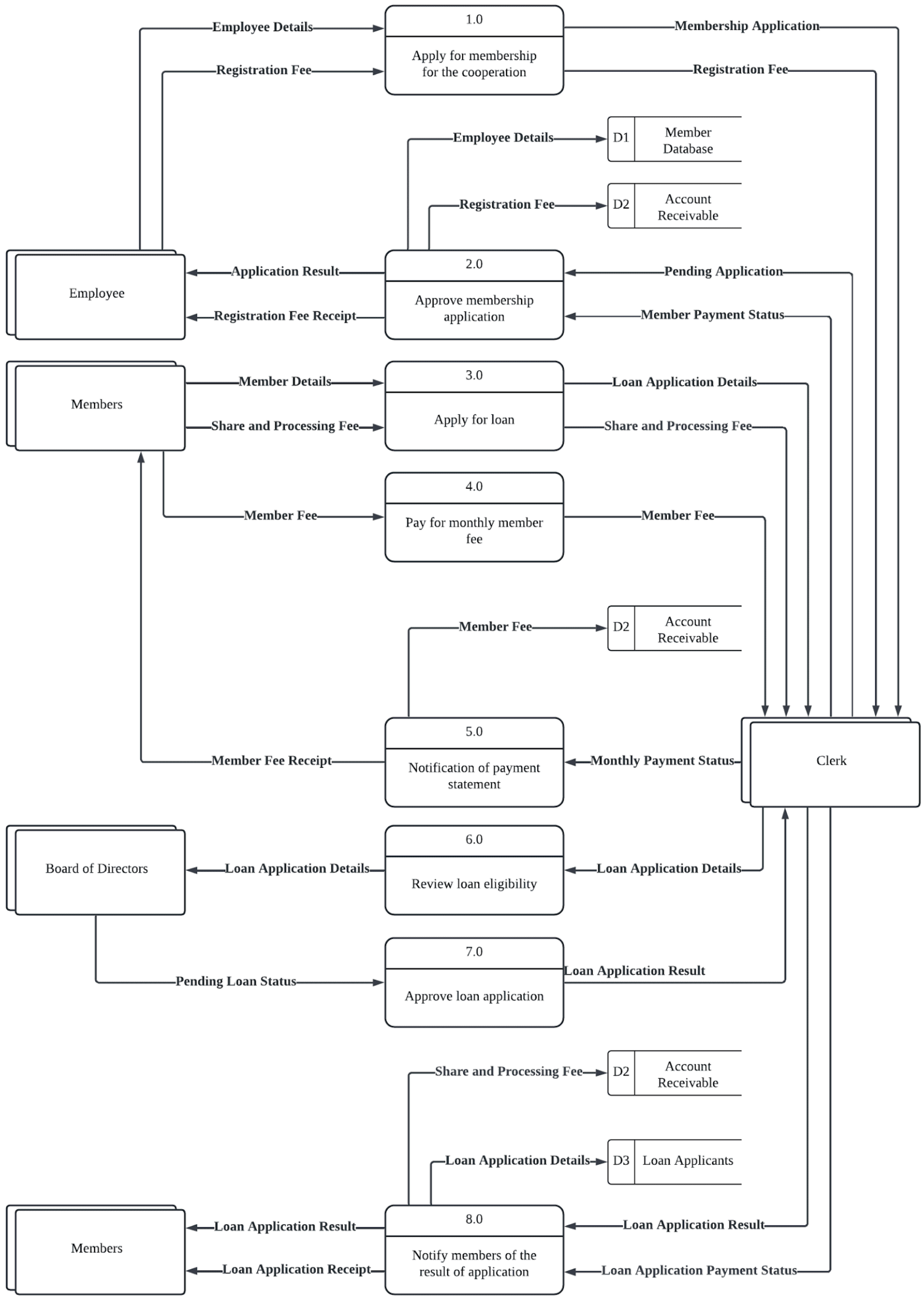


Figure 4.2 Diagram 0 of KADA's current Cooperative System

4.3 SYSTEM REQUIREMENTS

System requirements define the basis on which software will function. They include a wide range of topics, including outputs, inputs, processes, performance, and controls.

4.3.1 FUNCTIONAL REQUIREMENTS

- An employee of KADA is able to apply for membership by membership applications and pay the member fee.
- An employee is able to be notified of the result of a membership application by the clerk if they apply for a membership.
- A member is able to apply for loans by loan applications and pay the share and processing fee.
- A member is able to be notified of the result of a loan application by the clerk if they apply for a loan application
- The board of directors is able to view and approve loan applications.
- The clerk is able to receive information and process it before handing it to other parties in the system.
- The clerk is able to process fees and generate receipts for each payment.

4.3.2 NON-FUNCTIONAL REQUIREMENTS

- The system is able to operate during office hours only
- The system is able to store sensitive information safely.
- The system is able to comply with data regulation laws.
- The system is easy to understand and use.

4.4 SYSTEM DESIGN

4.4.1 PROPOSED SYSTEM

<u>CONTEXT DIAGRAM</u>

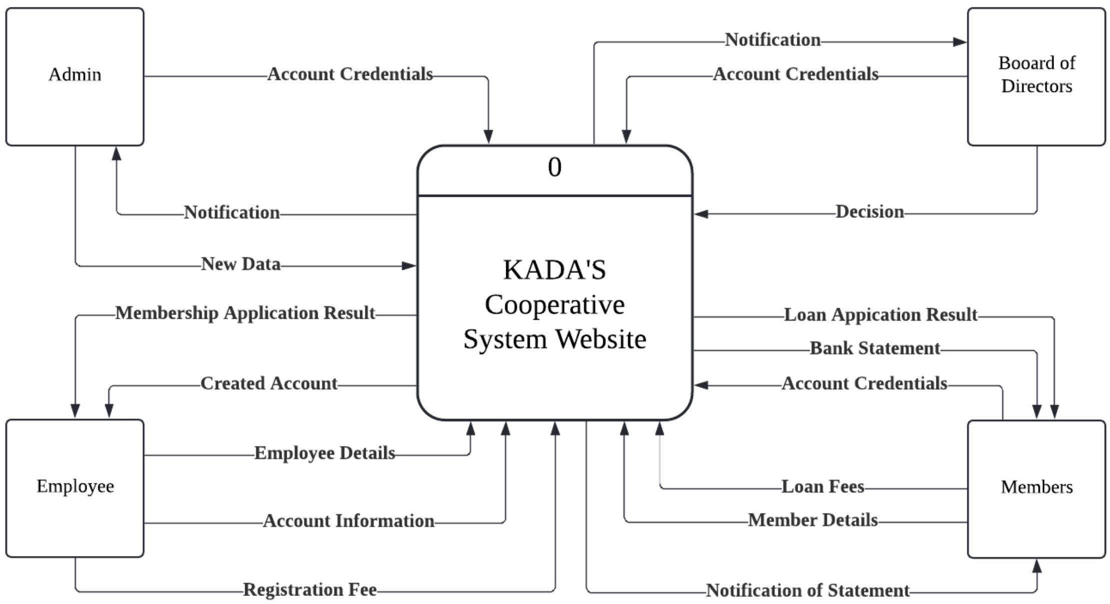


Figure 4.3 Logical Context Diagram of KADA's Cooperative System Website

Figure 4.3 Illustrates KADA's Cooperative Website System in the simplest form. It shows the data flow in and out from each entity showing how each entity interacts with the system. Data flowing in indicates the data that is provided by the entity to the system to undergo any related processes. Then, the output of the data flows from the system to the entities showing the result of the data provided from before.

LEVEL 0 DIAGRAM

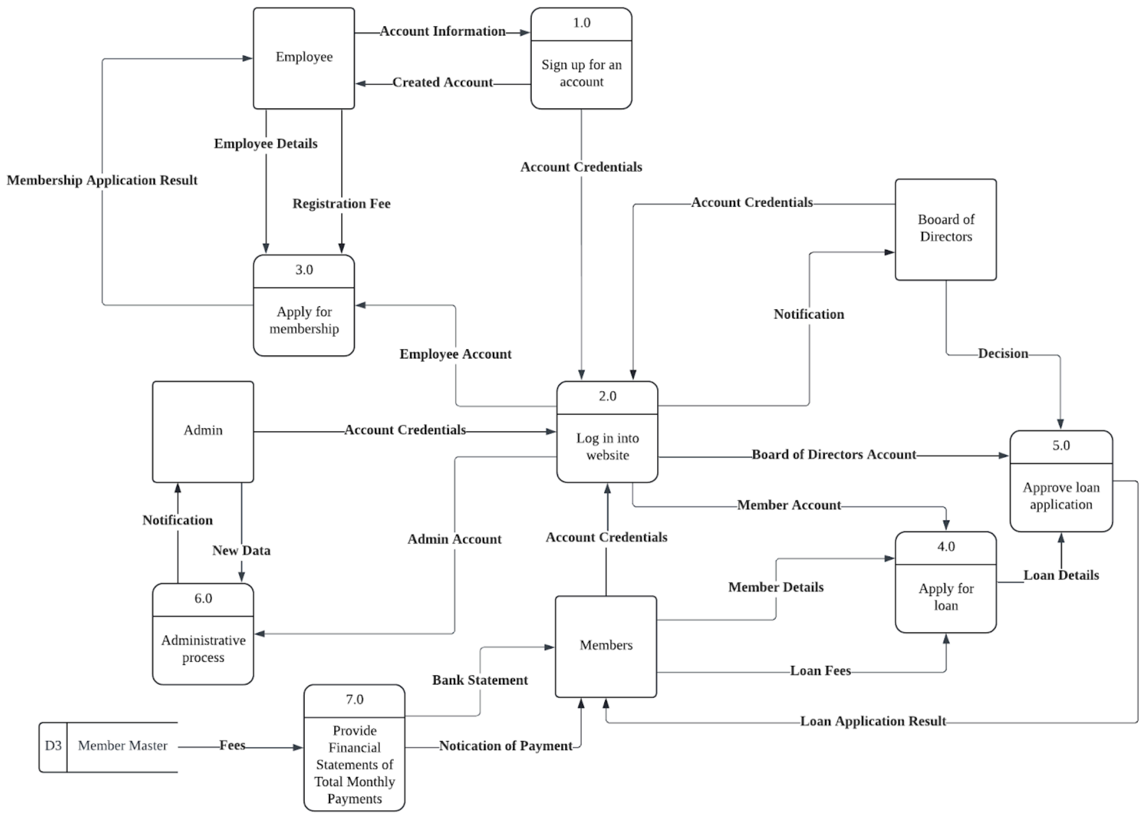


Figure 4.4 Level 0 Diagram KADA's Cooperative System

Figure 4.4 is a Level 0 Diagram that illustrates the system in a more detailed way. The diagram specifies all of the processes that exist within the system. Data flows from the entities that act as the input for the processes that might be involved in completing a certain task. If needed the data flows from one process to the other before an output of the task is sent back to the entities. Some processes require two entities before the final outcome of the task can be outputted. An example of that is the loan application process where it needs the approval (Process 5.0) of the Board of Directors for the loan applications (Process 4.0) by the Members.

CHILD DIAGRAM OF PROCESS 1.0

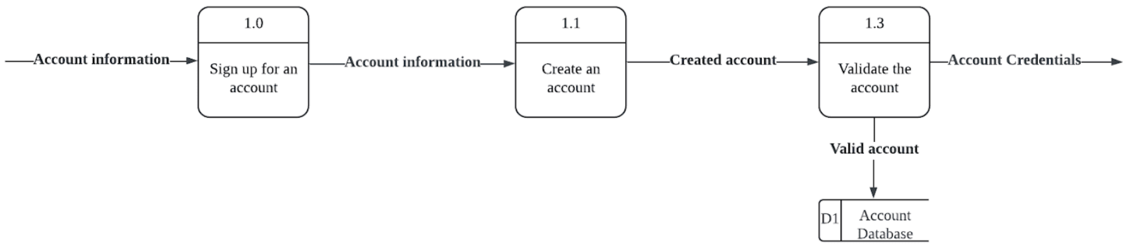


Figure 4.5 Child Diagram for Process 1.0

The child diagram from *Figure 4.5* represents a sequence of steps involved in creating and validating a user account. It starts with the user providing account information to sign up for an account (Process 1.0). Once the account information is submitted, it proceeds to the creation of the account (Process 1.1). After the account is created, the system moves to validate the account (Process 1.3). Upon successful validation, the account credentials are provided to the user, and the validated account is stored in the Account Database (D1). The diagram illustrates the flow of account information from initial sign-up through to validation and storage, ensuring that the account is properly created and authenticated before use.

CHILD DIAGRAM OF PROCESS 2.0

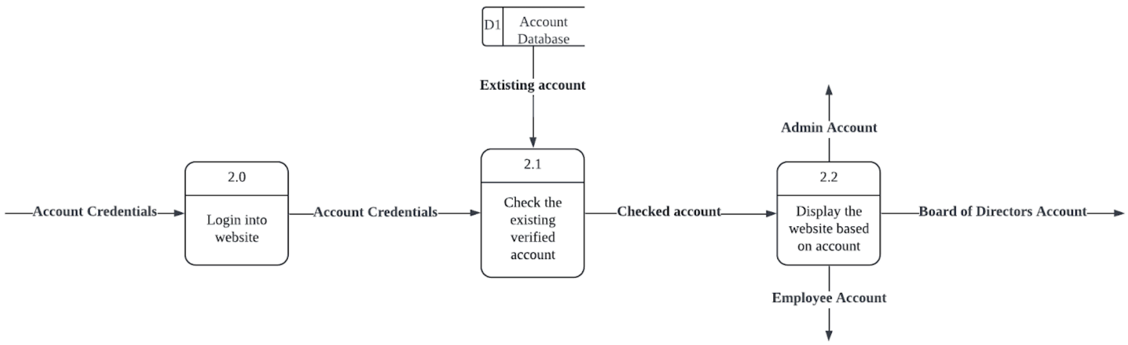


Figure 4.6 Child Diagram for Process 2.0

The child diagram in *Figure 4.6* outlines the process for logging into a website and displaying the appropriate interface based on the user's account type. The process begins with the user providing their account credentials to log into the website (Process 2.0). The system then checks these credentials against the existing verified accounts stored in the Account Database (D1) to ensure they are valid (Process 2.1). Once the account is verified, the system proceeds to display the website interface based on the type of account the user holds, such as an Admin Account, Employee Account, or Board of Directors Account (Process 2.2).

CHILD DIAGRAM OF PROCESS 3.0

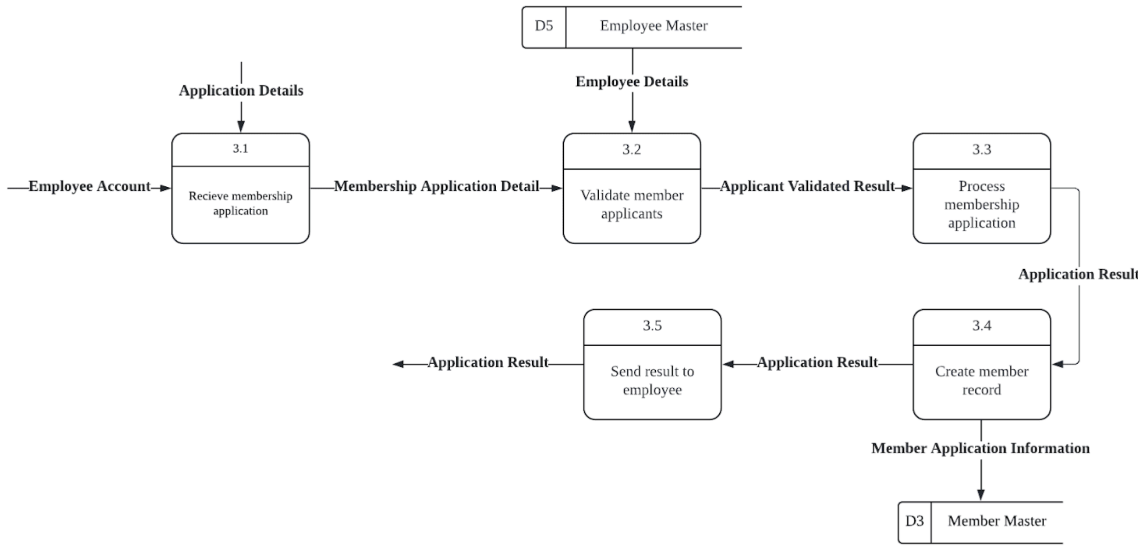


Figure 4.7 Child Diagram for Process 3.0

Figure 4.7 illustrates the process of handling membership applications. It begins with the process of receiving the membership application from an employee (Process 3.1), which outputs membership application details. These details are then validated using employee information (Process 3.2) from the Employee Master (D5). Upon validation, the application is processed (Process 3.3), resulting in an application result that is used to create a member record in the Member Master (Process 3.4). The final step is sending the application result back to the employee (Process 3.5). Data stores involved include the Employee Master, containing employee details used for validation, and the Member Master (D3), which stores information about all members. The flow of data makes sure that membership applications are efficiently received, validated, processed, recorded, and communicated back to the applicants.

CHILD DIAGRAM FOR PROCESS 4.0

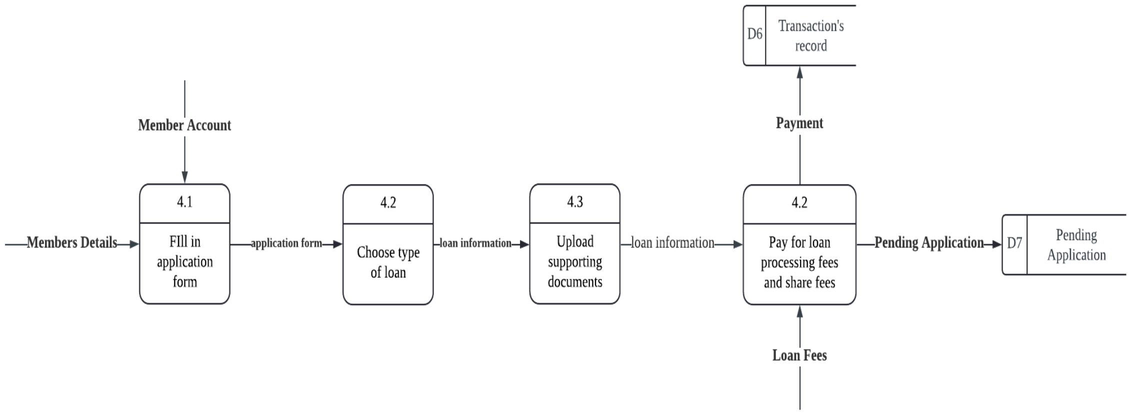


Figure 4.8 Child Diagram for Process 4.0

The child diagram of *Figure 4.8* illustrates a loan application process. It begins with members submitting their application form (4.1) along with their details and loan fees. After submitting the form, they proceed to pay for loan processing and share fees (4.2). The payment is recorded in the transaction's record (D6), and the application is marked as pending (D7).

CHILD DIAGRAM FOR PROCESS 5.0

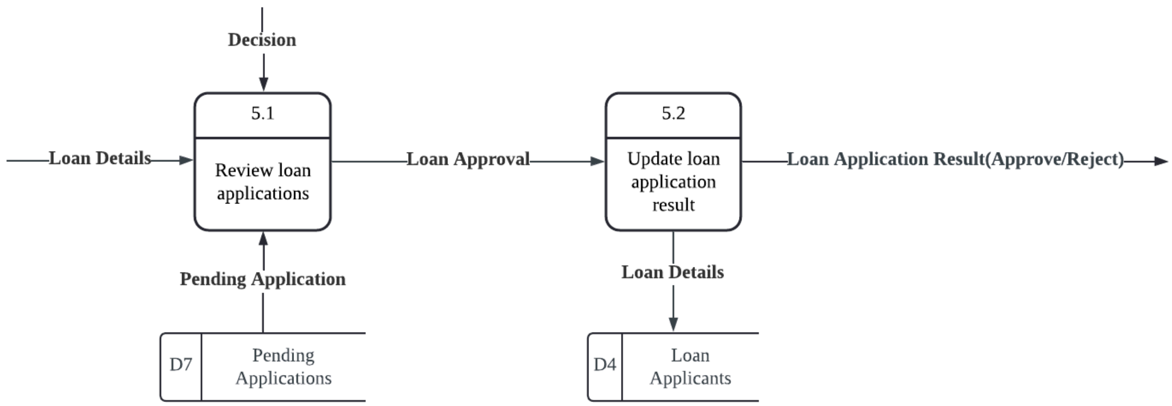


Figure 4.9 Child Diagram for Process 5.0

The child diagram in *Figure 4.9* explains the sub-processes involved in reviewing and updating loan applications, specifically highlighting processes 5.1 and 5.2.

In process 5.1, the loan details are received from the input. The review process involves evaluating the loan application and making a decision, which determines whether the application is approved or remains pending. If the application is pending, it is stored in the "Pending Applications" database (D7). Once the loan application is approved, it moves to process 5.2, "Update loan application result." This process involves updating the application status to reflect the decision made in the review stage. The updated loan application result is then recorded and communicated to the applicant. The relevant loan details are stored in the "Loan Applicants" database (D4), ensuring that all information is up-to-date.

CHILD DIAGRAM FOR PROCESS 6.0

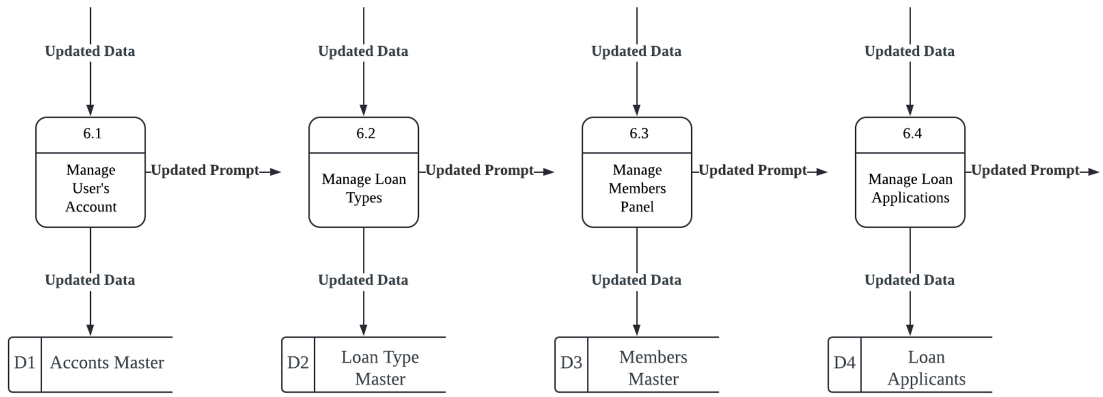


Figure 4.10 Child Diagram for Process 6.0

The child diagram in *Figure 4.10* describes the processes and data storage involved in a system that manages user accounts, loan types, members, and loan applications. The diagram includes four main processes: managing user accounts (Process 6.1), managing loan types (Process 6.2), managing the members panel (Process 6.3), and managing loan applications (Process 6.4). Each of these processes receives updated data inputs, processes this data by updating, adding, or deleting records, and stores the results in corresponding master data stores. The data stores are namely Accounts Master (D1), Loan Type Master (D2), Members Master (D3) and Loan Applicants Master (D4). Each process also produces updated prompts for subsequent updates to indicate successful data updates.

CHILD DIAGRAM FOR PROCESS 7.0

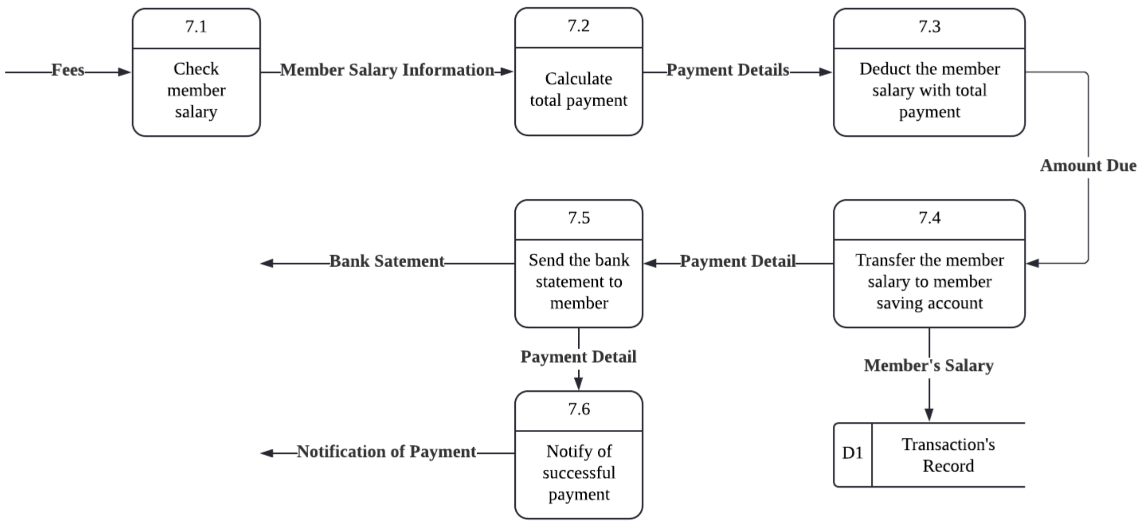


Figure 4.11 Child Diagram for Process 7.0

The child diagram in *Figure 4.11* explains the process of handling member salary payments. It begins with checking the member's salary (Process 7.1) and then calculating the total payment (Process 7.2). After calculating the payment details, the member's salary is adjusted accordingly (Process 7.3). The adjusted salary is then transferred to the member's savings account (Process 7.4). The bank statement is sent to the member (Process 7.5), and a notification of successful payment is issued (Process 7.6). The member's salary transaction is recorded in the transaction's record (D1). The flow includes interactions between the member's salary information, payment details, and bank statements to ensure proper tracking and notification.

PROCESS SPECIFICATION

Process 1.3

```
IF Created Account has the same information as existing user
    DO WHILE Created Account has the same information as existing user
        Create another account with different information
    ENDDO
ELSE
    Update the account information into the database
    Output Account Credential for Login Process
ENDIF
```

Process 2.1

```
DO
READ Account Database for Existing Accounts
BEGIN IF
IF Account Username is correct
    DO WHILE Password is not correct
    Check Password
    ENDDO
    Verify Login
ELSE
    Enter Username again
END IF
```

Process 2.3

```
DO
READ Type of account
IF Account type is Administrator
    Display Administrator Account Interface
ELSE IF Account type is Board of Directors
    Display Board of Directors Account Interface
ELSE IF Account type is Members
    Display Members Account Interface
ELSE IF Account type is Employee
    Display Employee Account Interface
END IF
```

Process 3.2

```
IF Employee applied is not an Employee at KADA
        Reject Application
ELSE
        Move to next process
ENDIF
```

Process 4.2

```
DO Pay loan processing fee and share fee
IF loan processing fee < RM50 and share fee < RM300
        Reject loan application
ELSE
        Update transaction record
        Update pending application
ENDIF
```

Process 5.2

```
IF Loan is approved
        Update loan applicants database
        Delete pending application database
        Output successful loan application
ELSE
        Delete pending application database
        Output failed loan application
```

Process 6.1

```
IF New data is new user

        Add new user record

ELSE IF New data is updated user record

        Update existing user record

ELSE IF  New data is non-active user

        Delete user record

ENDIF
```

Process 6.2

IF New data is new loan type

Add new loan type record

ELSE IF New data is updated user record

Update existing loan type record

ELSE IF New data is non-active user

Delete loan type record

ENDIF

Process 6.3

IF New data is new members record

Add new member record

ELSE IF New data is updated member record

Update existing member record

ELSE IF New data is non-active member

Delete member record

ENDIF

Process 6.4

IF New data is new loan applicant

Add new loan applicant record

ELSE IF New data is updated loan applicant record

Update existing loan applicant record

ELSE IF New data is settled loan payment

Delete loan applicant record

ENDIF

Process 7.2

IF Member has loan application
Total payment = loan repayment + member fee
ELSE
Total payment = member fee
ENDIF

4.4.2 PROPOSED SYSTEM : PHYSICAL DFD (TO BE) SYSTEM

LEVEL 0 DIAGRAM

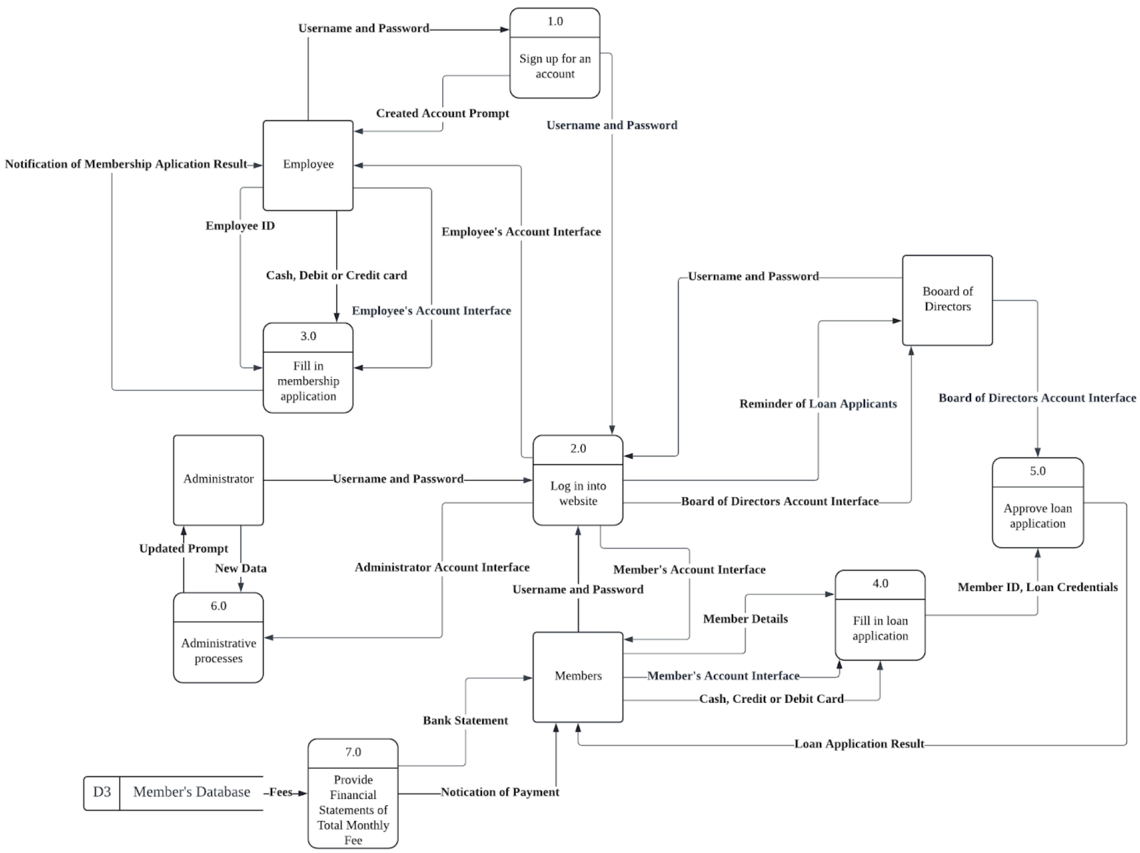


Figure 4.12 Level 0 Diagram

Figure 4.12 is a Level 0 of a Physical Data Flow Diagram that illustrates the system more detailed than the logical data flow diagram. The diagram depicts how the system will be implemented. The details of all of the processes and the data flows to be more specific. It also represents manual processes, programs and program modules and also databases used in the system. For example, the data flow for “Fees” as an input for process 3.0 is changed to “Credit or Debit Card” that specifies the payment method for that process. This also applies to other processes if needed.

CHILD DIAGRAM OF PROCESS 1.0

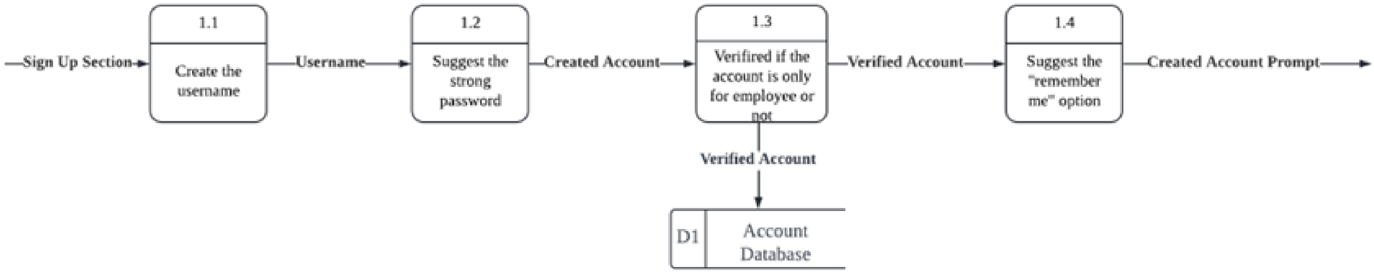


Figure 4.13 Child Diagram for Process 1.0

This diagram from *Figure 4.13* details the account creation and verification process within a sign-up section. It begins with the user creating a username (Process 1.1). Following this, the system suggests a strong password for the user (Process 1.2). Once the account is created, it moves to the verification phase where the account is verified by checking whether the account is from the KADA or not. This will prevent the unauthorised accounts from logging into the system (Process 1.3). The verified account information is then stored in the Account Database (D1). Finally, the system suggests the "remember me" option to the user for easier future access (Process 1.4).

CHILD DIAGRAM OF PROCESS 2.0

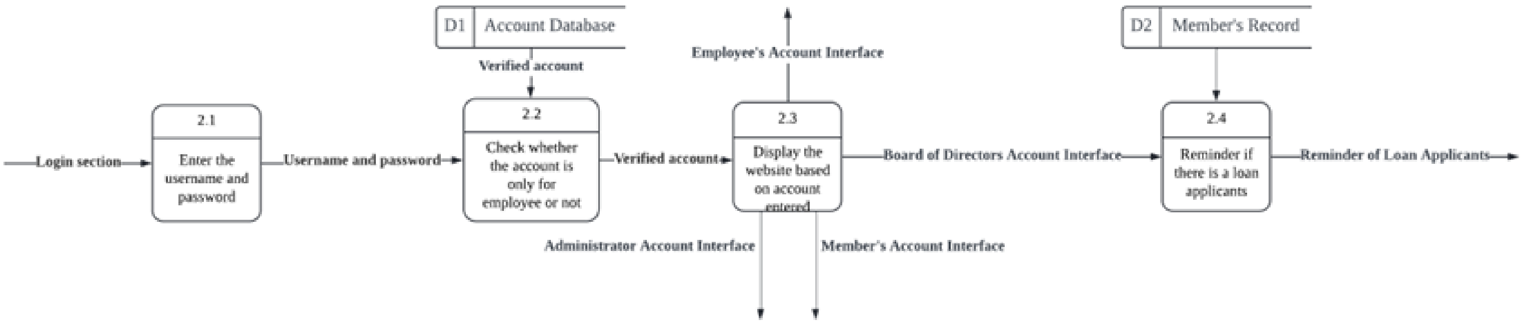


Figure 4.14 Child Diagram for Process 2.0

This diagram from *Figure 4.14* illustrates the workflow of a login and account verification system. Users start by entering their username and password (Process 2.1). The system then checks the entered credentials against the account database (D1) to verify the account and also checks if the account has only been assigned from KADA (Process 2.2). If the account is verified, different interfaces are displayed based on the type of account: Administrator, Employee, Member, or Board of Directors (Process 2.3). For Members, the system additionally checks the Member's Record (D2) and provides reminders if there are any pending loan applications (Process 2.4).

CHILD DIAGRAM FOR PROCESS 3.0

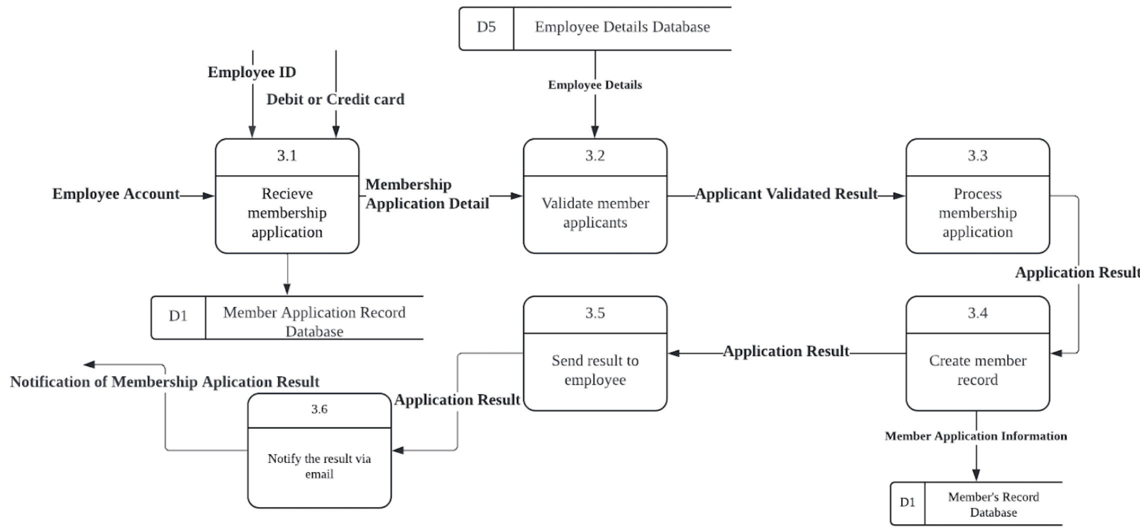


Figure 4.15 Child Diagram for Process 3.0

This child diagram in *Figure 4.15* describes the process of handling membership applications for employees. It starts with receiving the membership application (Process 3.1) along with the employee ID and payment information (debit or credit card). The application details are then validated against the employee details database (Process 3.2). Once validated, the membership application is processed (Process 3.3). If successful, a member record is created (Process 3.4). The result of the application is then sent to the employee (Process 3.5), and the member application record is updated. Finally, the employee is notified of the application result via email (Process 3.6). The flow ensures proper verification, processing, and notification of membership applications, maintaining records in both the member application record database and the member's record database.

CHILD DIAGRAM FOR PROCESS 4.0

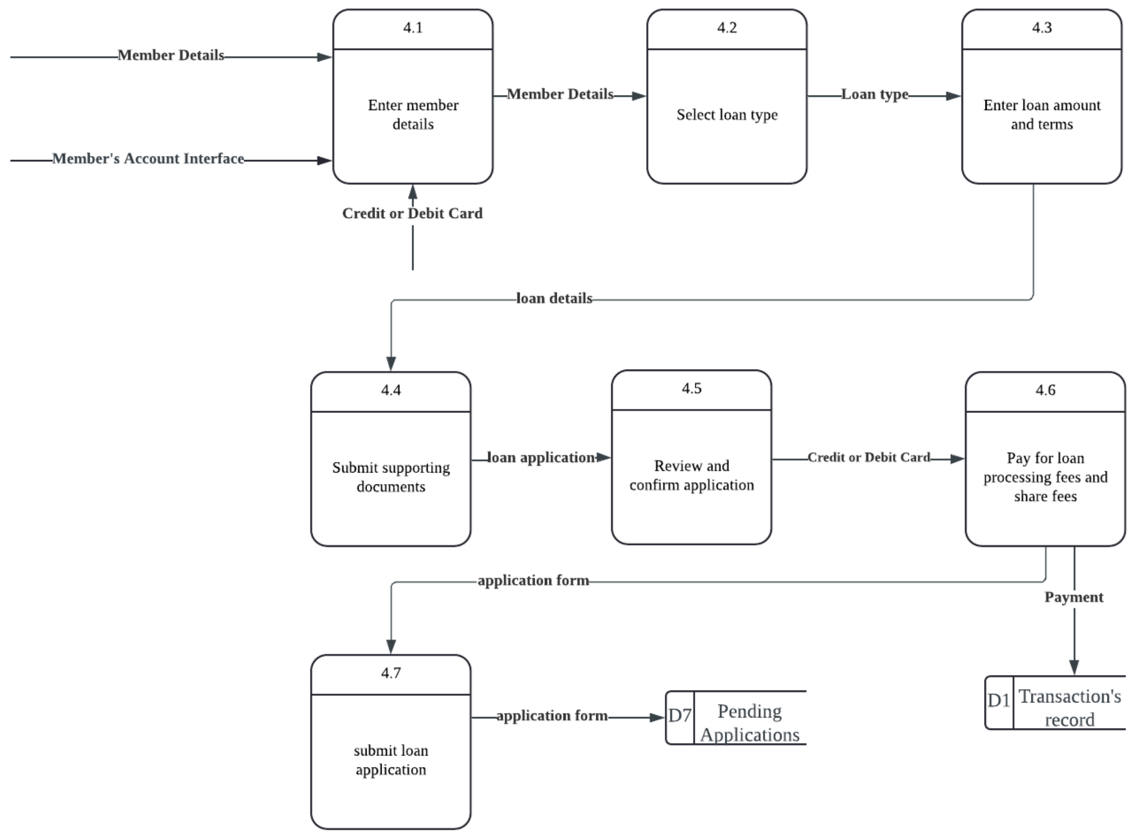


Figure 4.16 Child Diagram for Process 4.0

The child diagram for process 4.0 as shown in *Figure 4.16* explains the loan application process starting with the member entering their personal details through the Member's Account Interface. After providing their details, the member selects the type of loan they wish to apply for in process 4.2 and then specifies the loan amount and terms in process 4.3. Next, the member submits any required supporting documents for the loan application.

Following the submission of documents, the loan application is reviewed and confirmed to ensure all provided details and documents are accurate. Once the application is confirmed, the member proceeds to pay the loan processing fees and share fees using a credit or debit card. This payment is recorded in the transaction's record.

Finally, the completed loan application is submitted, moving it into a pending status for further processing. This structured process ensures that all necessary details are captured, reviewed, and the relevant fees are paid before the loan application is officially submitted.

CHILD DIAGRAM FOR PROCESS 5.0

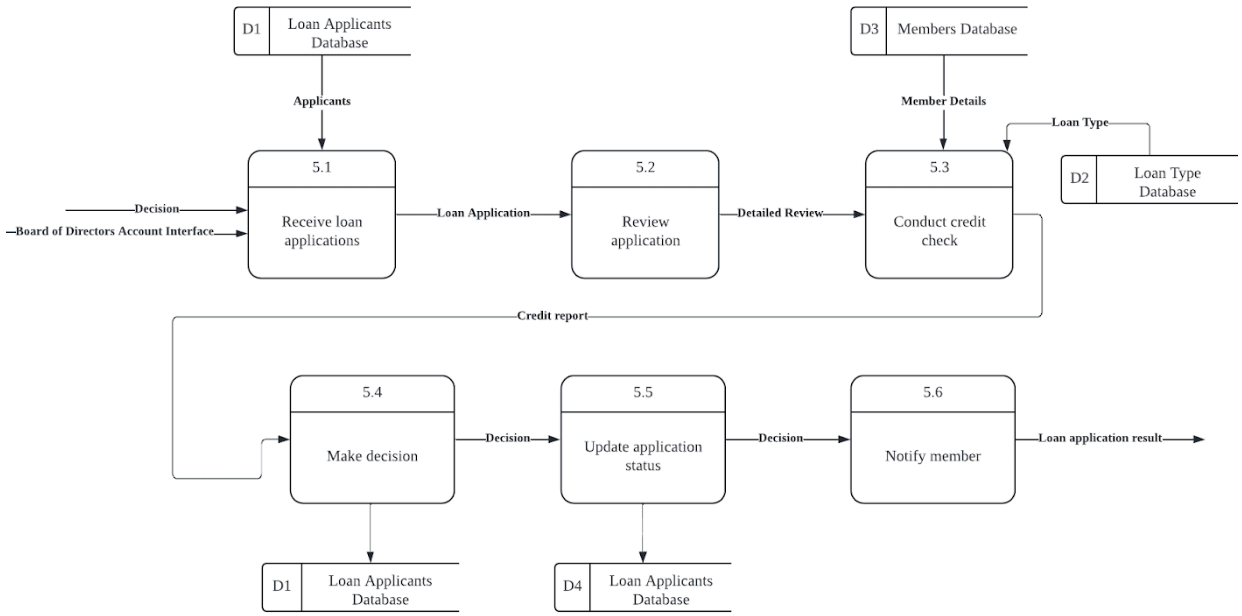


Figure 4.17 Child Diagram for Process 5.0

This child diagram for Process 5.0 in *Figure 4.17* shows the loan application review and approval process. It begins with receiving loan applications (5.1) from the Loan Applicants Database (D1). Once an application is received, it undergoes an initial review (5.2) to ensure all necessary information is provided.

The application then proceeds to a detailed review (5.3), where a credit check is conducted using data from the Members Database (D3) and the Loan Type Database (D2). Based on the credit report generated from the detailed review, a decision is made (5.4) regarding the application. The application status is then updated (5.5) in the Loan Applicants Database (D4) to reflect this decision. Finally, the member is notified (5.6) of the outcome of their loan application. This process ensures a thorough review of each application, considering the member's credit history and the type of loan requested before a final decision is made and communicated to the applicant.

CHILD DIAGRAM FOR PROCESS 6.0

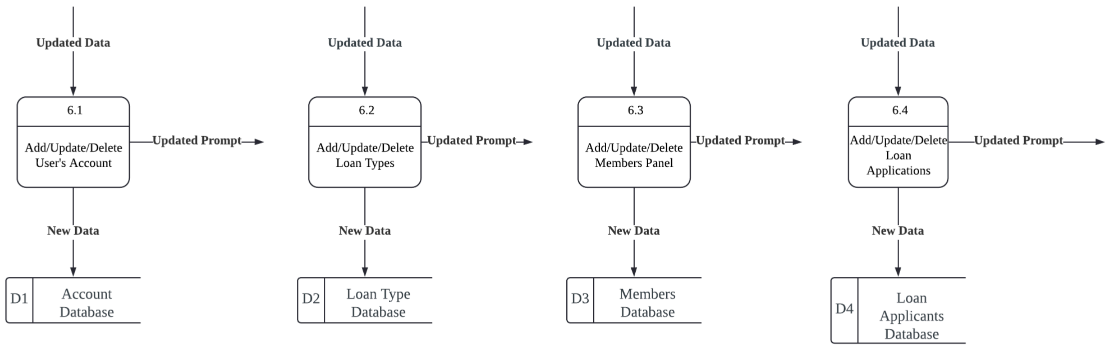


Figure 4.18 Child Diagram for Process 6.0

The child diagram in *Figure 4.18* provides a detailed visualisation of the sub-processes of the Administrative process (Process 6.0) by an administrator that handles user accounts, loan types, members, and loan applications. It consists of four main processes: adding, updating, and deleting user accounts (Process 6.1), loan types (Process 6.2), members (Process 6.3), and loan applications (Process 6.4). Each process receives updated data, processes it by either adding, updating, or deleting records, and then stores this data in their respective databases namely the Account Master (D1), Loan Type Master (D2), Member Master (D3), Loan Applicants (D4).

CHILD DIAGRAM FOR PROCESS 7.0

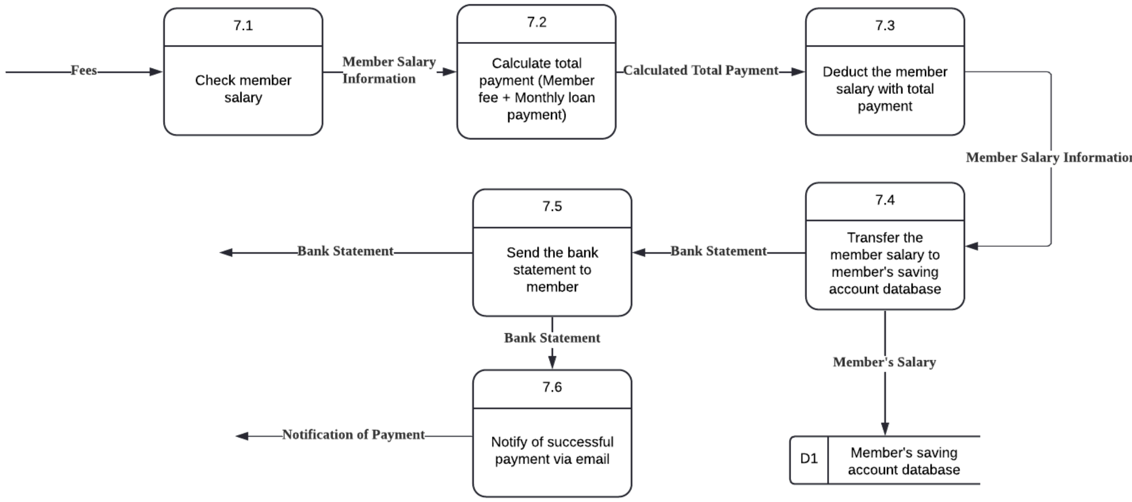


Figure 4.19 Child Diagram for Process 7.0

This child diagram in Figure 4.19 describes the process of managing member salary deductions and transfers. The process starts with checking the member's salary (Process 7.1). The total payment is then calculated, which includes the member fee and any monthly loan payment (Process 7.2). This calculated total payment is used to deduct the corresponding amount from the member's salary (Process 7.3). The remaining salary is then transferred to the member's saving account database (Process 7.4). Subsequently, a bank statement is generated and sent to the member (Process 7.5), and a notification of successful payment is sent via email (Process 7.6). The entire process ensures accurate deductions, transfers, and notifications, and maintains records in the member's saving account database (D1).

4.4.3 PARTITIONING

Partitioning is the process of analysing a data flow diagram and deciding how to break it into collections of manual methods and computer programmes. There are several reasons to perform partitioning including two user groups, timing, similar tasks, efficiency, consistency of data and security.

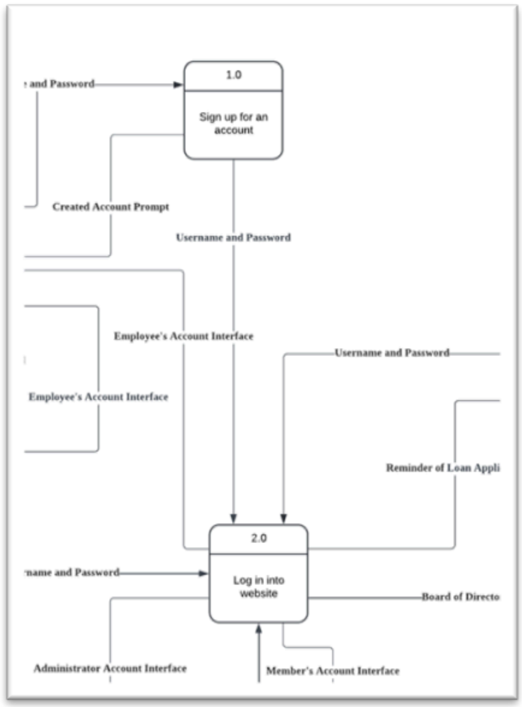


Figure 4.20 Partitioning for Process 1.0 &2.0

Figure 4.20 indicates process 1 and process 2 are partitioned together because they are dependent on each other. After the user successfully signs up for their respective account, they for sure need to log in to the website to ensure that their account has been verified or not. The only difference is that the signup process just needs to register it once, while the login process can be done multiple times.

Figure 4.21 Partitioning for Process 3.0 &4.0

Figure 4.21 indicates process 3 and process 4 have the almost same task which is to fill the Applications but with different user groups (employee and member). All employees can apply for membership but not necessarily. Meanwhile, for the loan application, only members can apply for it but not employees. These two are totally different user groups and have their own advantages.

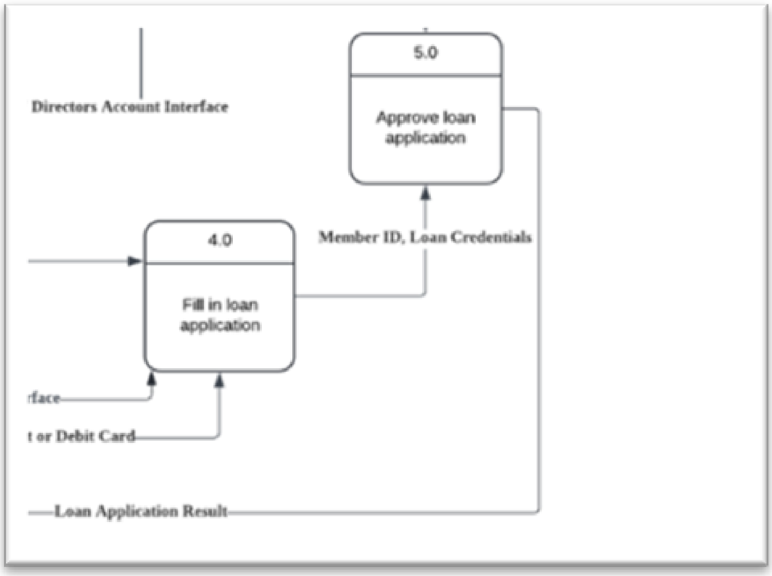


Figure 4.22 Partitioning for Process 4.0 & 5.0

Figure 4.22 indicates process 4 and process 5 can be partitioned together because they depend on each other. These two processes must be next to each other, after a member fills up the application then the Board of Directors must approve or decline to continue the next step which the application results in.

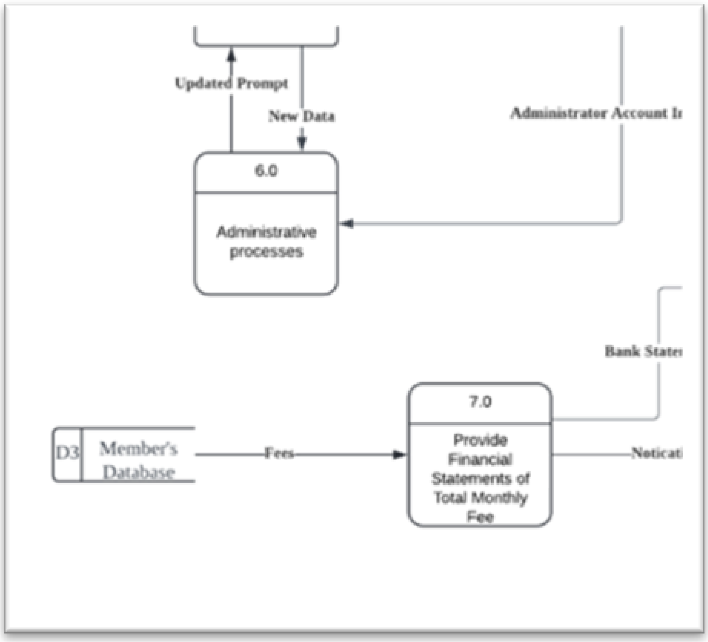


Figure 4.23 Partitioning for Process 6.0 & 7.0

In the *Figure 4.23* process number 6 and 7 are both a different process but must be partitioned because they are related to each other. Process 6 is used to update the data store while process 7 will read the data store.

4.4.4 CRUD MATRIX

A CRUD matrix is a very effective tool when it comes to the capturing and displaying of activities and permission within a system. In turn, it is very useful to add the CRUD Matrix with the analysis of the user processes in the context of the actors and roles who perform the processes within the system to complete the picture. The acronym CRUD is often used for Create, Read, Update, and Delete, the activities that must be present in a system for each master file. For example, when doing Employee Login activity, it will read the Account Database which is one of the datastore in the system

Activity	Account Database	Loan Type	Member Database	Loan Applicants	Employee Database	Loan Application	Member Saving's Account
Employee Login	R						
Member Login	R						
Board of Directors Login	R						
Admin Login	R						
Apply for a Membership			C		R		
Apply for a Loan		R	R		R	C	
Approve Loan Application				C		R	
Change User Account	CRUD						
Change Loan Types	CRUD						
Change Loan Applicants	CRUD						
Change Member Details	CRUD						
Employee Sign Up	C				R		
Transfer Member Salary			R				U

Table 4.24 CRUD matrix

4.4.5 EVENT RESPONSE TABLE

Four events are used to construct 4 event response tables. The events are members applying for loans, employees applying for membership, the Board of Directors approving loan applications, and the administrator doing administrative processes. The tables are illustrated below.

EVENT: MEMBER APPLY FOR LOAN

Event	Source	Trigger	Activity	Response	Destination
Member log in	Member	Member username and password	Find member records in the database and verify password. Send Welcome Web page.	Welcome Web page	Member
Member fill in loan application form	Member	Member information	Find membership records in the database and verify the process.	Member loan application Data stored	Member
Member confirm the application	Member	Member confirmation	Send confirmation to the server.	The confirmation page (accept or reject)	Board of director
Member pay for loan processing fees and share fees	Member	Member payment method	Payment sent to database.	Payment transaction checking	Board of director
Members submit the application.	Member	Member submission	Application information sent to database.	Submission confirmation	Board of director
Member receive the approval	Board of director	Board of director approval	The approval letter was sent to the member via email.	Notification in member device	Member

Table 4.25 Event Response Table for event members applying for loan

EVENT: EMPLOYEE APPLIES FOR MEMBERSHIP

Event	Source	Trigger	Activity	Response	Destination
Employee login website	Employee	Employee username and password	Find employee record and verify password. Send Welcome Web Page	Welcome Web page	Employee
Employee do a member application	Employee	Application details	List the application requirement	Employee information	Admin
Application submission	Employee	Application details	Send the application. Display the submission	Submitted application	Admin
Eligibility check	Admin	Application details	Verify the application eligibility	Information checking	Admin
Approval/denial decision	Admin	Click “Approve” or “Reject” button	Display the employee application. Store data on Member Record	Decision approval	Admin
Send employee email	Admin	Temporal	Send employee an email application decision	Notification sent to the employee	Employee
Membership Fee	Employee	Deduct from salary or pay away	Display the fee	Fee displayed	Employee

Table 4.26 Event Response Table for event employee applies for membership

EVENT: BOARD OF DIRECTORS APPROVE LOAN APPLICATIONS

Event	**Source**	**Trigger**	**Activity**	**Response**	**Destination**
Receive Loan Application	Board of Directors	Receive application	Loan application details are received from the member	Loan application stored in Loan Applicants Database (D1)	Board of Directors
Review Application	Board of Directors	Check loan details	BOD reviews the loan application.	Evaluation report generated	Board of Directors
Conduct credit check	Board of Directors	BOD decision	Perform a creditworthiness assessment of the member	Credit report generated and reviewed	Board of Directors
Make decision	Board of Directors	Approve or reject	BOD decides to approve or reject the loan application	Decision (approve/reject)	Board of Directors
Update application status	Board of Directors	Update to applicant	Update the loan application status based on the decision	Application status updated in Loan Applicant Database(D1)	Loan Applicants Database(D1)
Notify member	Board of Directors	Notification to member	Notify the member of the board's decision	Notification sent to member	Member

Table 4.27 Event Response Table for Board of Directors approve loan application

EVENT: ADMINISTRATOR DOES ADMINISTRATIVE PROCESSES

Event	Source	Trigger	Activity	Response	Destination
Administrator logs on	Administrator	Administrator ID and password	Find administrator record and verify password. Send Welcome Web page.	Welcome Web page	Administrator
Administrator browses Accounts Database, Loan Types Database, Members Database and Loan Applicants	Administrator	New Data	Views and reads all the records that exist in the database system.	System's Database Page	Administrator
Administrator adds new information in the database	Administrator	Click's "Add" button in the database	Adds new records/data to the database	Updated Data Prompt	Administrator
Administrator updates existing information in the database	Administrator	Click's "Update" button in the database	Updates existing records/data in the database	Updated Data Prompt	Administrator
Administrator updates existing information in the database	Administrator	Click's "Delete" button in the database	Deletes existing records/data in the database	Updated Data Prompt	Administrator

Table 4.28 Event Response Table for event administrator does administrative processes

4.4.6 STRUCTURE CHART

To visualize the steps involved in Process 1 to 7, we developed a structure chart. A structure chart is a hierarchical diagram that breaks down a system into its component processes. It illustrates the flow of information and control among them. This chart helps in understanding the functional components of the process and their interactions to ensure a systematic and clear representation of the workflow.

<u>PROCESS 1.0</u>

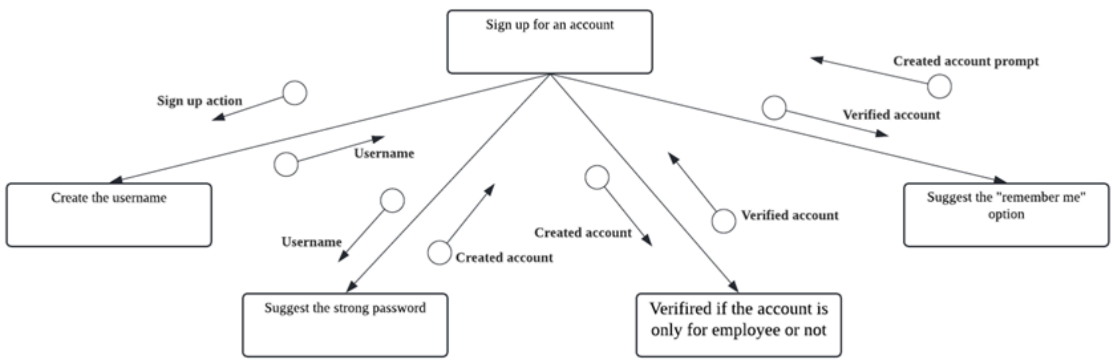


Figure 4.29 Structure Chart for Process 1.0

The diagram in *Figure 4.29* outlines the process for signing up for an account. It begins with the main action "Sign up for an account" at the centre, branching out to several steps. First, users are prompted to create a username, during which the system may suggest actions like verifying the created account and suggesting a strong password. Once the account is created, it must be verified, often through the account only for the employee. After verification, users might see a prompt to save their login information with a "remember me" option. Each step in the process ensures the creation of a secure and verified user account.

PROCESS 2.0

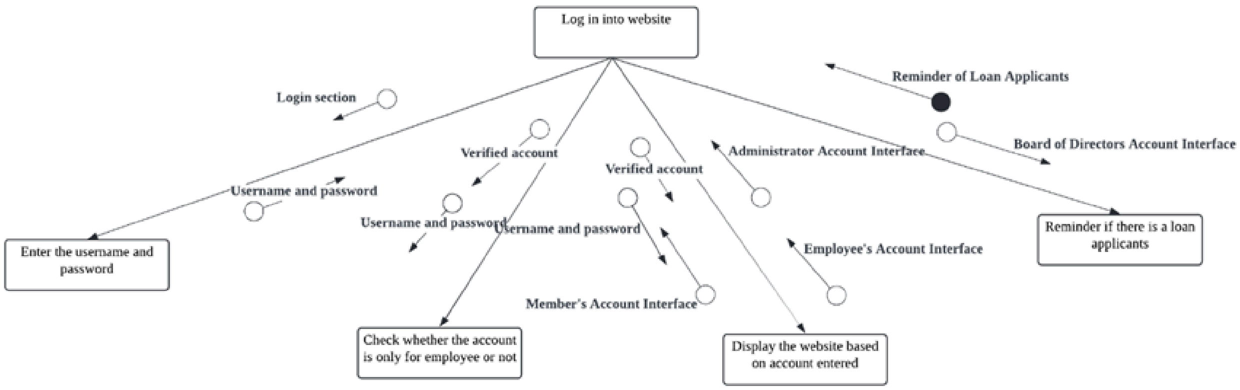


Figure 4.30 Structure Chart for Process 2.0

The diagram in *Figure 4.30* outlines the process of logging into a website. It starts with the user entering their username and password in the login section. The system then checks whether the account is verified and also only for the employee. If the account is verified, the user is directed to the appropriate interface based on their role: Administrator, Employee, Member, or Board of Directors. For members, there is an additional step where the system checks for any loan applications and provides reminders if there are any. This ensures that users access the correct interface and receive relevant information based on their account status and role.

PROCESS 3.0

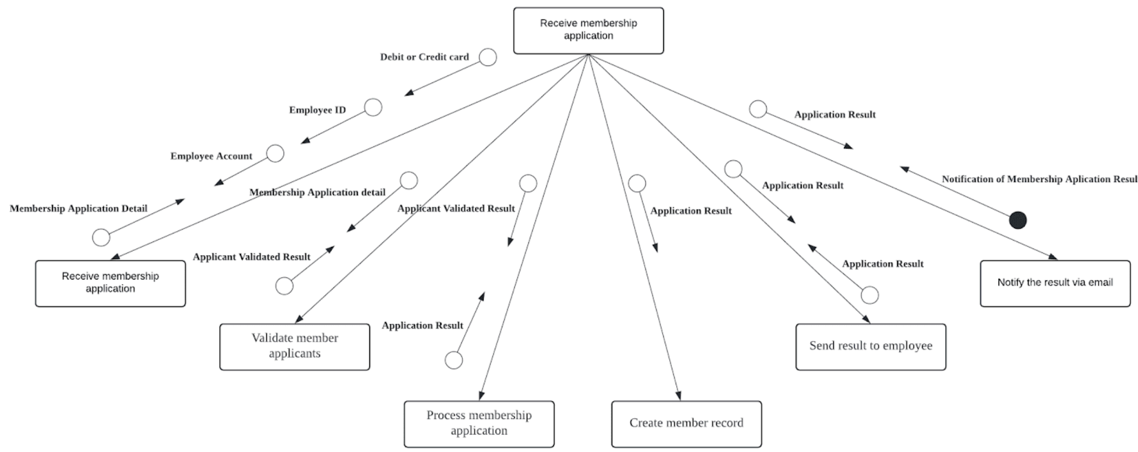


Figure 4.31 Structure Chart for Process 3.0

This structure chart in *Figure 4.31* explains the process of handling membership applications. It starts with receiving the membership application, which then undergoes validation to ensure the applicants meet the necessary criteria. Following validation, the application is processed, and a member record is created. The results of the application (whether approved or denied) are then sent to the relevant employee account and also to the applicant via email. The diagram shows multiple data flows, including membership application details, validation results, application results, and notification of the application results, ensuring all relevant parties are informed at each step of the process.

PROCESS 4.0

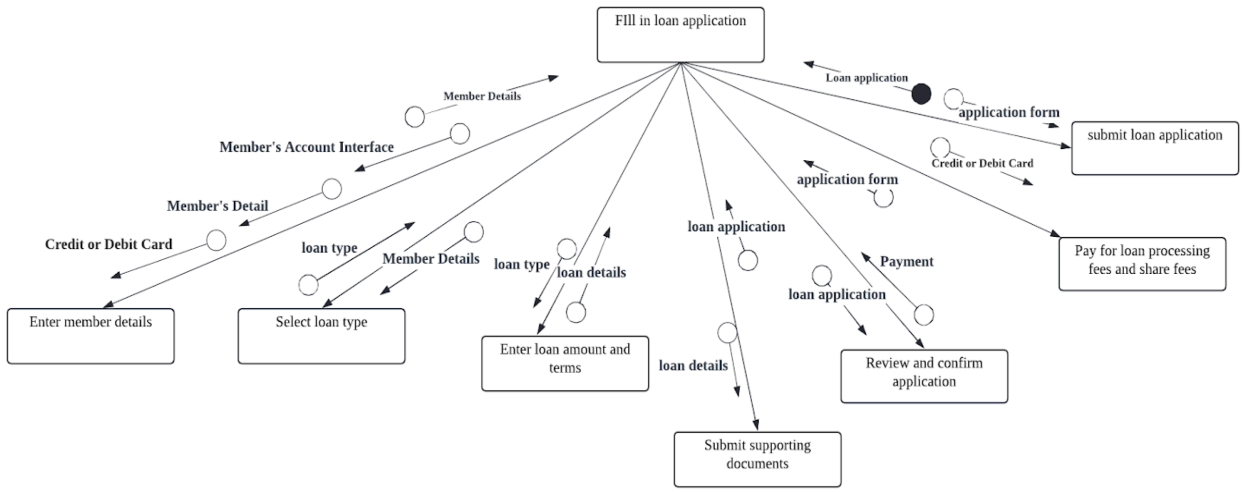


Figure 4.32 Structure Chart for Process 4.0

The structure chart as in *Figure 4.32* is for the fill-in application process that starts with entering member details, which can be accessed via the member's account interface. Next, the applicant selects the loan type and then inputs the loan amount and terms. Member details and loan-type information are required in this step. Supporting documents are then submitted as part of the loan application process. After these inputs, the applicant reviews and confirms the application, which includes checking loan details and confirming the loan application. The final steps involve paying for loan processing fees and share fees using credit, or debit cards, followed by submitting the loan application form. Each step in the process is interconnected, ensuring that all necessary information is collected and verified before the final submission.

PROCESS 5.0

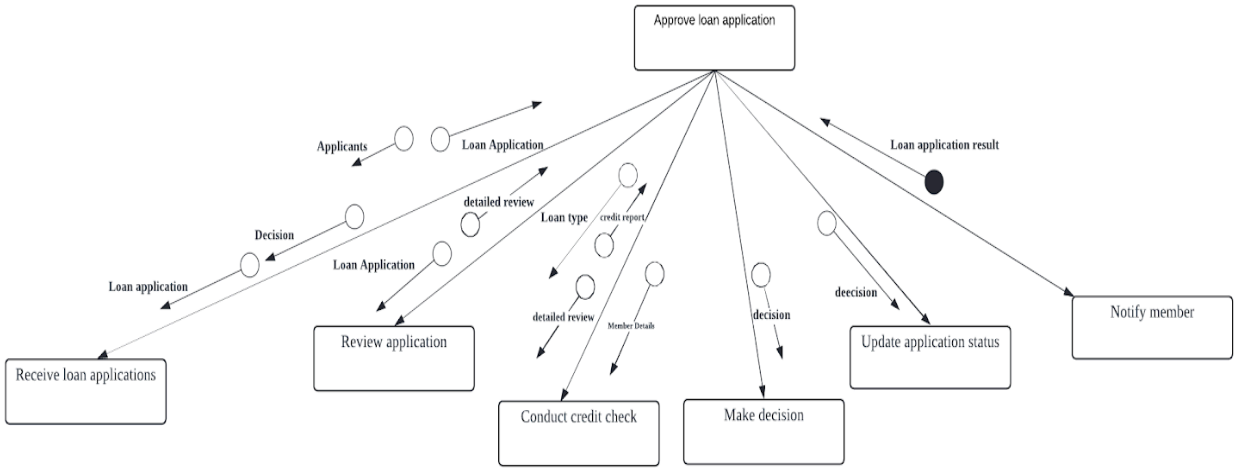


Figure 4.33 Structure Chart for Process 5.0

Figure 4.33 explains the structure chart for Process 5.0 and all the steps involved in approving a loan application. The process begins with the reception of loan applications, which then undergo a review stage. During the review, a detailed examination of the application is conducted, and applicants' information is verified. Then, a credit check is performed to examine the applicant's creditworthiness based on loan type, credit report, and member details.

After conducting the credit check, a decision is made by the Board of Directors regarding the approval or rejection of the loan application. The decision along with detailed review results are used for the next step, which is to update the application status. The result of the loan application is then informed to the member in completing the process.

PROCESS 6.0

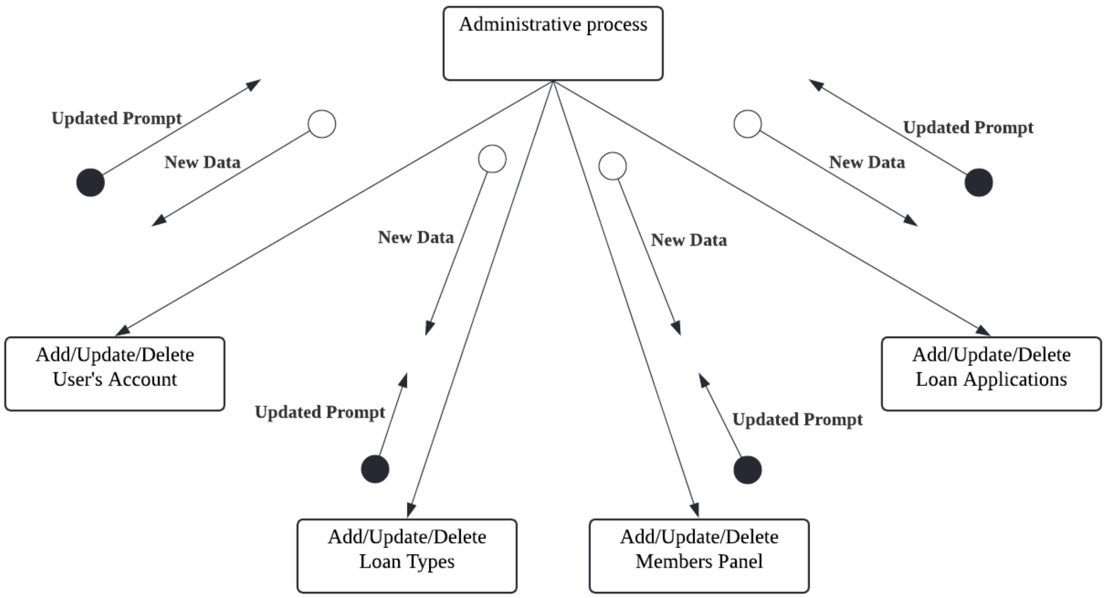


Figure 4.34 Structure Chart for Process 6.0

In *Figure 4.34,* This structure chart depicts an administrative process overseeing multiple functionalities: amendments as to the creation of new ones, modification or even deletion of the user profiles, loan application types, loan types, and the membership panels. New data is provided to the central administrative process by these functionalities where it is analyzed, and in return the updated prompts are provided. Every function (for instance, Add/Update/Delete User's Account) both supplies information to the central administrative process and derives signals from it, making all changes consistent within the system.

PROCESS 7.0

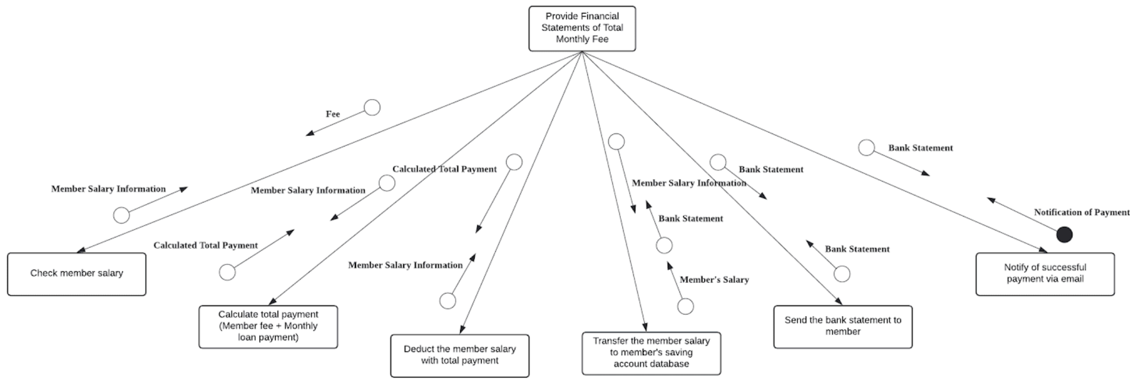


Figure 4.35 Structure Chart for Process 7.0

This structure chart in *Figure 4.35* illustrates the process of providing financial statements of total monthly fees for members. It begins with checking the member's salary, which then feeds into the calculation of the total payment, including member fees and monthly loan payments. This calculated total payment is used to deduct the member's salary accordingly. The deducted salary information is transferred to the member's savings account database. Following this, a bank statement is generated and sent to the member, along with an email notification confirming the successful payment. The entire process is centralized around the goal of providing accurate financial statements.

4.4.7 SYSTEM ARCHITECTURE

A system architecture is a conceptual model that outlines the structure, behaviour, and other aspects of a system. An architecture description is a formal description and representation of a system organised in such a way that it allows for reasoning about the system's structures and behaviours. Figure below shows the system architecture of our website.

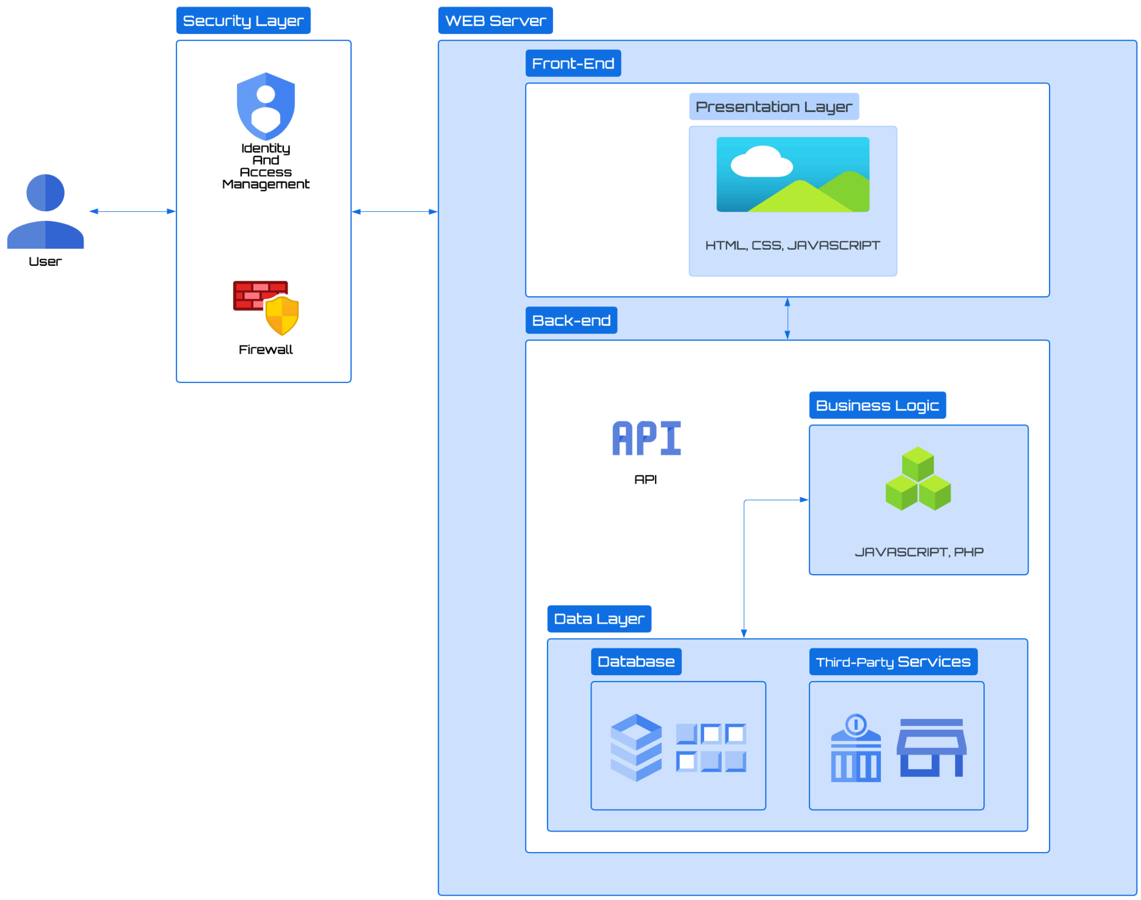


Figure 4.36 System Architecture of KADA's Cooperative Website System

The system is divided into multiple layers for easy illustration. First, the user will log into the system. For this process, security is the top priority to ensure no data leaks occur. In the security layer, there will be an identity and access management system where the user type will be identified. This step is necessary as different users will have their own corresponding interfaces. For example, only a verified Board of Directors account will have the power to approve loans in the web system. Next, the firewall is hardware or software put in a computer system that will check all the traffic going into or out of the computer or a network and decide whether or not to let the traffic through depending on a set of rules.

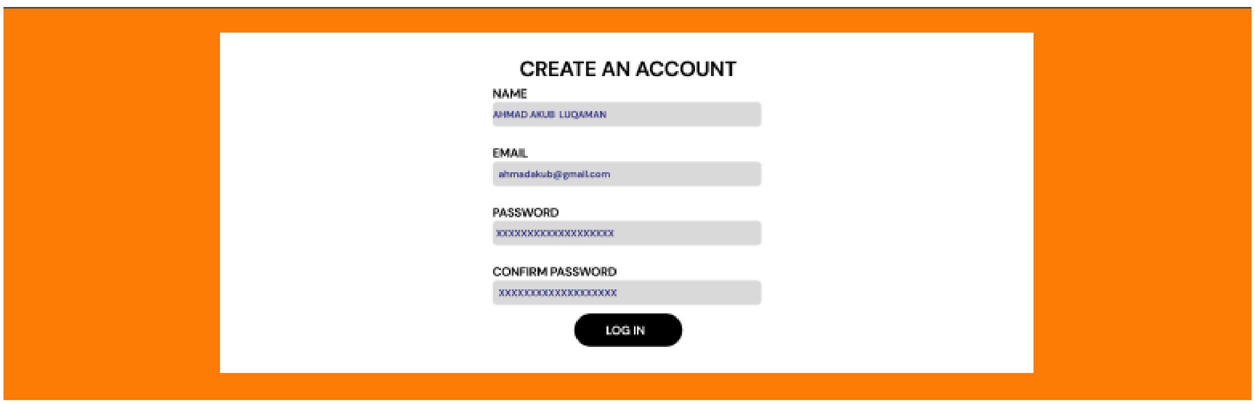
After that is the Web Server which consists of two components which are the front-end and back-end of the website. The front end consists of a presentation layer that shows what the user sees on the screen. For this layer, HyperText Markup Language (HTML), Cascading Style Sheets (CSS) and Javascript will be used to build the layer.

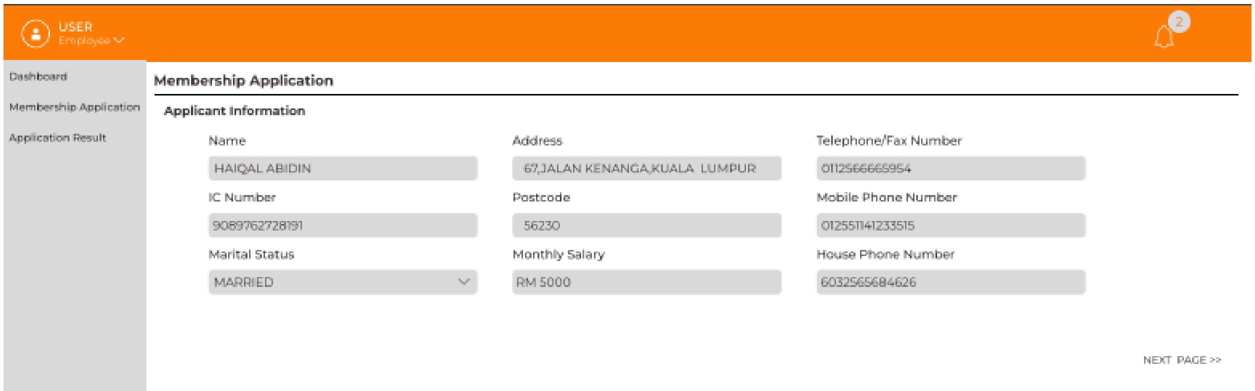
Finally is the back end of the website. It consists of a business logic layer and a data layer. The business logic is the business-specific programs or procedures governing the flow of data between a database and a user interface. This will be built using Javascript and PHP. The data layer will consist of a database, where all the data are stored and third-party services such as banks and other external systems will be used. All of the layers will be held together by Application Programming Interface (APIs) that serve as a way for two or more computer programs or components to communicate with each other.

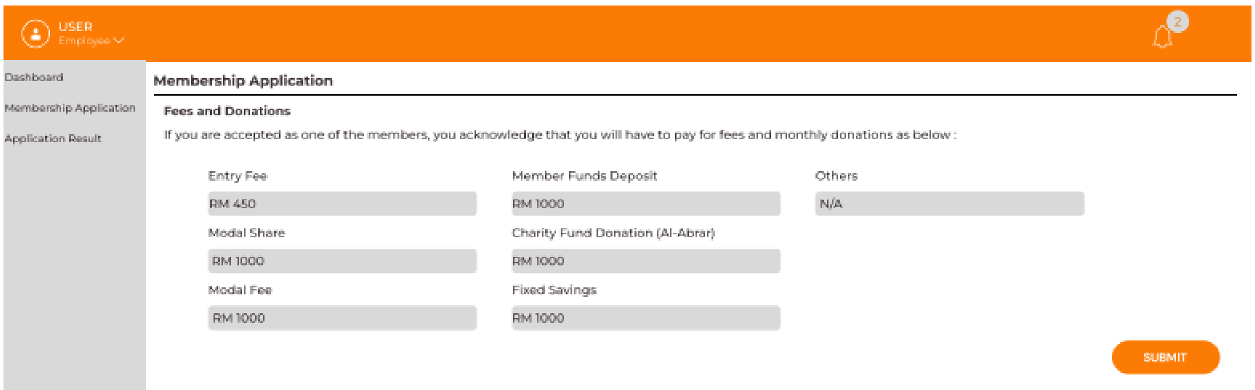
4.4.8 INTERFACE DESIGN (INPUT/OUTPUT DESIGN)

Figma Link :
https://www.figma.com/design/uImpDMlznmpifvYVONaJMU/Demo-Interface?node-id=0-1&t=zWLccxFdcMO6zDQO-1

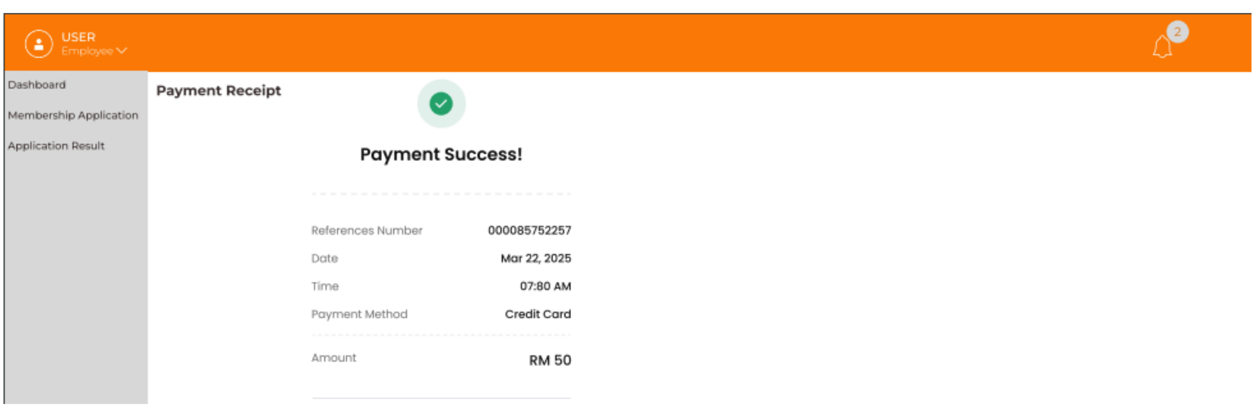
EMPLOYEE INTERFACE

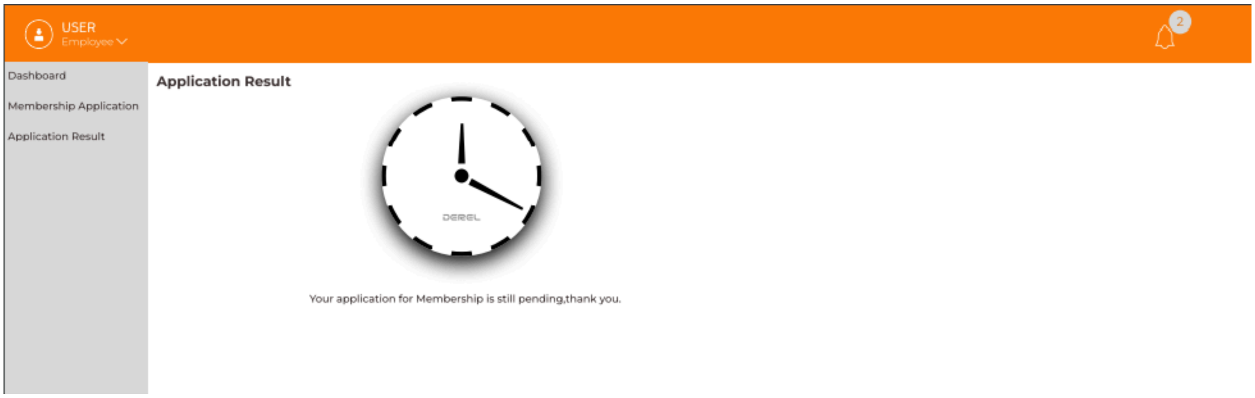



USER
Employee
2
Dashboard
Membership Application
Application Result
Membership Application
Applicant Information
Name
HAIQAL ABIDIN
Address
67,JALAN KENANGA,KUALA LUMPUR
Telephone/Fax Number
0112566665954
IC Number
9089762728191
Postcode
56230
Mobile Phone Number
012551141233515
Marital Status
MARRIED
Monthly Salary
RM 5000
House Phone Number
6032565684626
NEXT PAGE >>

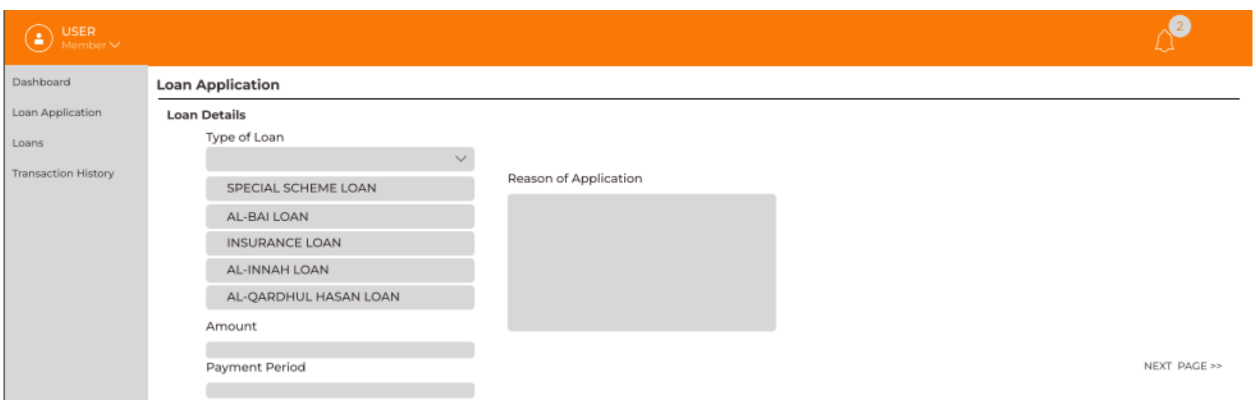

USER
Employee
2
Dashboard
Membership Application
Application Result
Membership Application
Fees and Donations
If you are accepted as one of the members, you acknowledge that you will have to pay for fees and monthly donations as below :
Entry Fee
RM 450
Member Funds Deposit
RM 1000
Others
N/A
Modal Share
RM 1000
Charity Fund Donation (Al-Abrar)
RM 1000
Modal Fee
RM 1000
Fixed Savings
RM 1000
SUBMIT

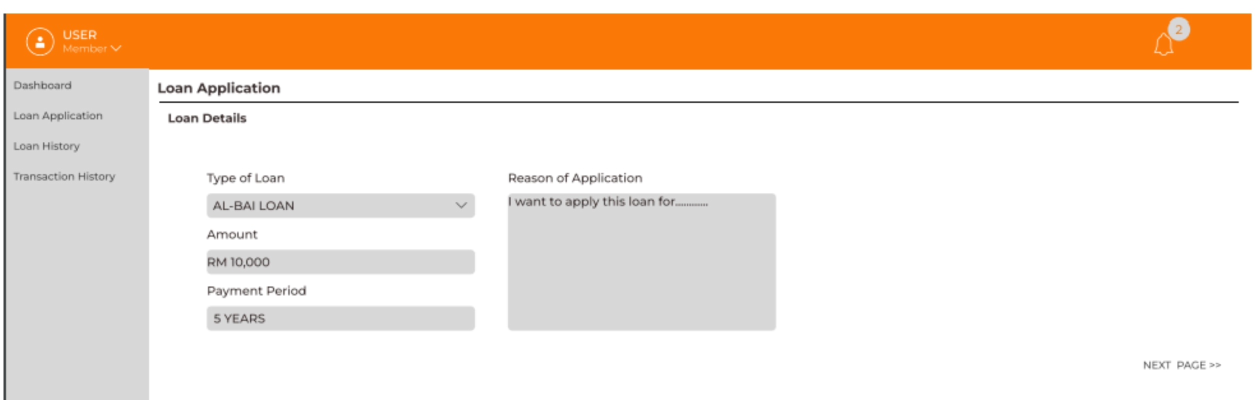

USER
Employee
2
Payment Gateway
Make payment using your Credit/Debit Card
VISA
PayPal
FPX
CIMB Clicks

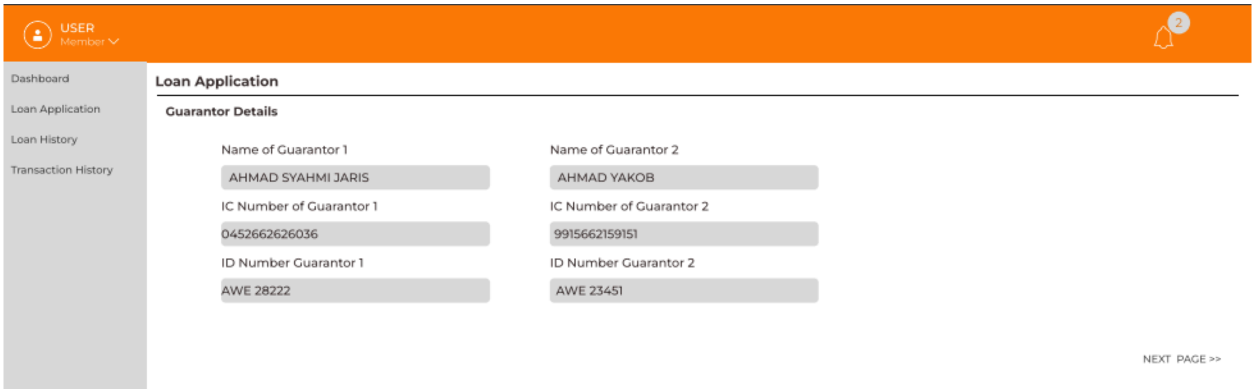

USER
Employee
2
Dashboard
Membership Application
Application Result
Payment Receipt
Payment Success!
References Number
000085752257
Date
Mar 22, 2025
Time
07:80 AM
Payment Method
Credit Card
Amount
RM 50

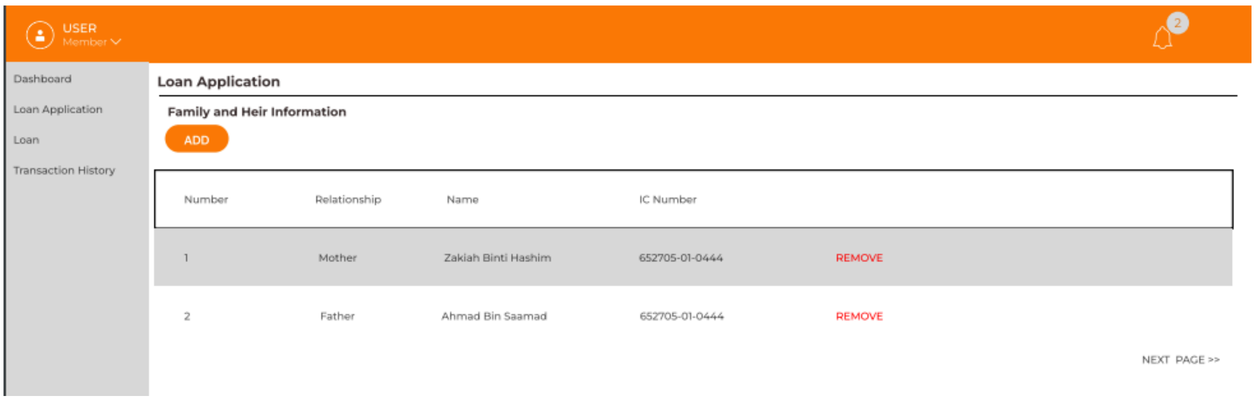

USER
Employee
2
Dashboard
Membership Application
Application Result
Application Result
DEREL
Your application for Membership is still pending,thank you.

MEMBER INTERFACE

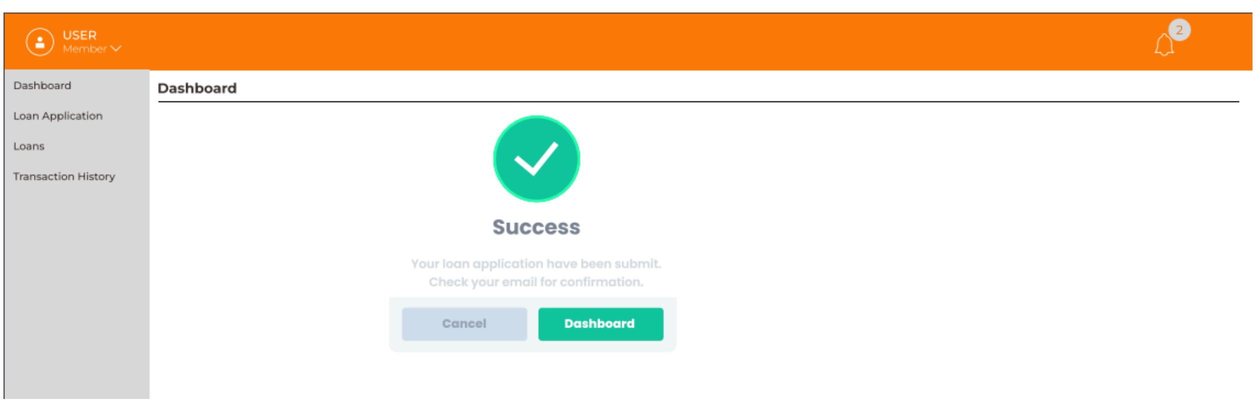


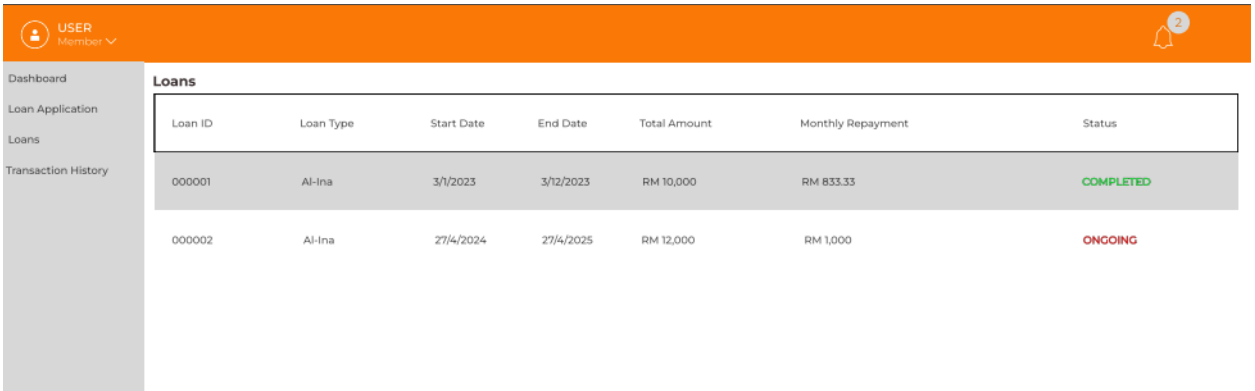


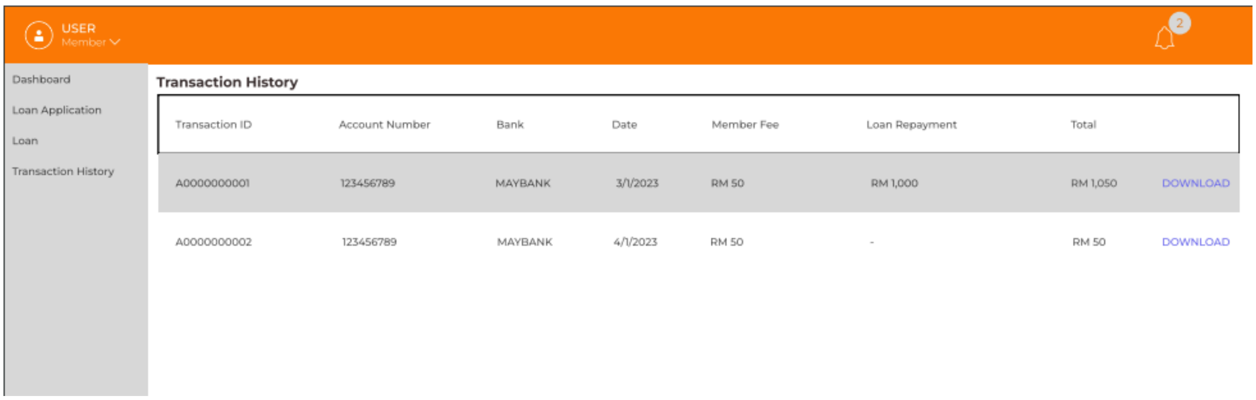

USER
Member
2
Dashboard
Loan Application
Loan History
Transaction History
Loan Application
Guarantor Details
Name of Guarantor 1
AHMAD SYAHMI JARIS
IC Number of Guarantor 1
0452662626036
ID Number Guarantor 1
AWE 28222
Name of Guarantor 2
AHMAD YAKOB
IC Number of Guarantor 2
9915662159151
ID Number Guarantor 2
AWE 23451
NEXT PAGE >>


USER
Member
2
Dashboard
Loan Application
Loan
Transaction History
Loan Application
Family and Heir Information
ADD
Number
Relationship
Name
IC Number
1
Mother
Zakiah Binti Hashim
652705-01-0444
REMOVE
2
Father
Ahmad Bin Saamad
652705-01-0444
REMOVE
NEXT PAGE >>

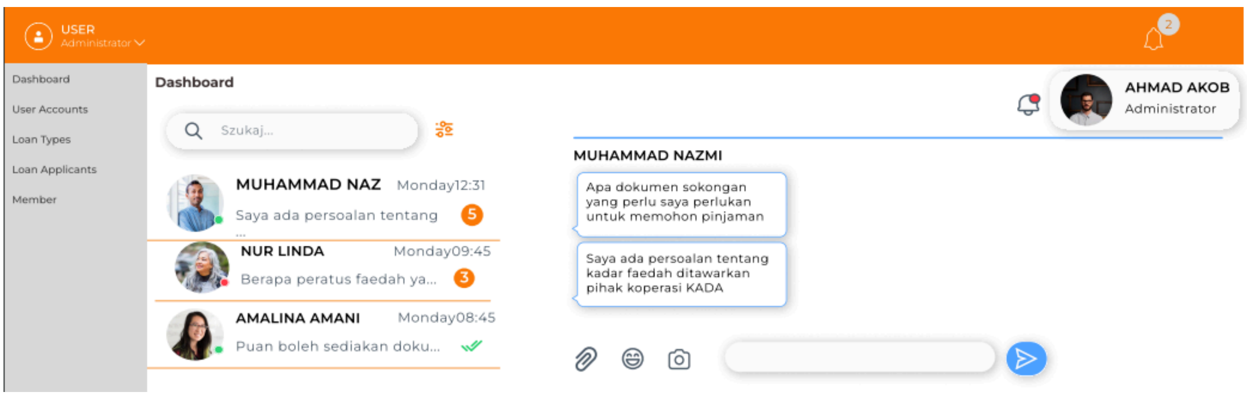

USER
Member
2
Dashboard
Loan Application
Loan History
Transaction History
Loan Application
Document Submission
Signature and Proof Guarantor 1
PDF
Signature and Proof Guarantor 2
PDF
Employer Validation
PDF
SUBMIT

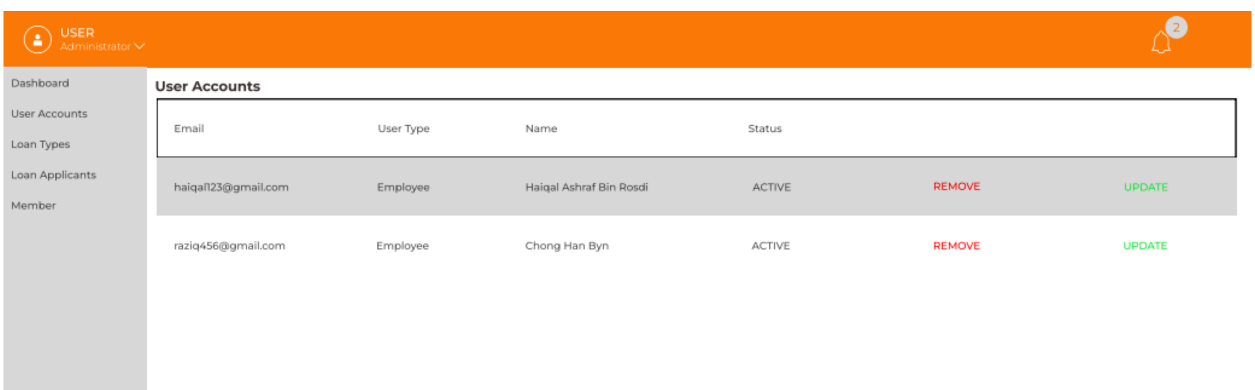

USER
Member
2
Dashboard
Loan Application
Loans
Transaction History
Dashboard
Success
Your loan application have been submit.
Check your email for confirmation.
Cancel
Dashboard


USER
Member
2
Dashboard
Loan Application
Loans
Transaction History
Loans
Loan ID
Loan Type
Start Date
End Date
Total Amount
Monthly Repayment
Status
000001
Al-Ina
3/1/2023
3/12/2023
RM 10,000
RM 833.33
COMPLETED
000002
Al-Ina
27/4/2024
27/4/2025
RM 12,000
RM 1,000
ONGOING


USER
Member
2
Dashboard
Loan Application
Loan
Transaction History
Transaction History
Transaction ID
Account Number
Bank
Date
Member Fee
Loan Repayment
Total
A0000000001
123456789
MAYBANK
3/1/2023
RM 50
RM 1,000
RM 1,050
DOWNLOAD
A0000000002
123456789
MAYBANK
4/1/2023
RM 50
-
RM 50
DOWNLOAD

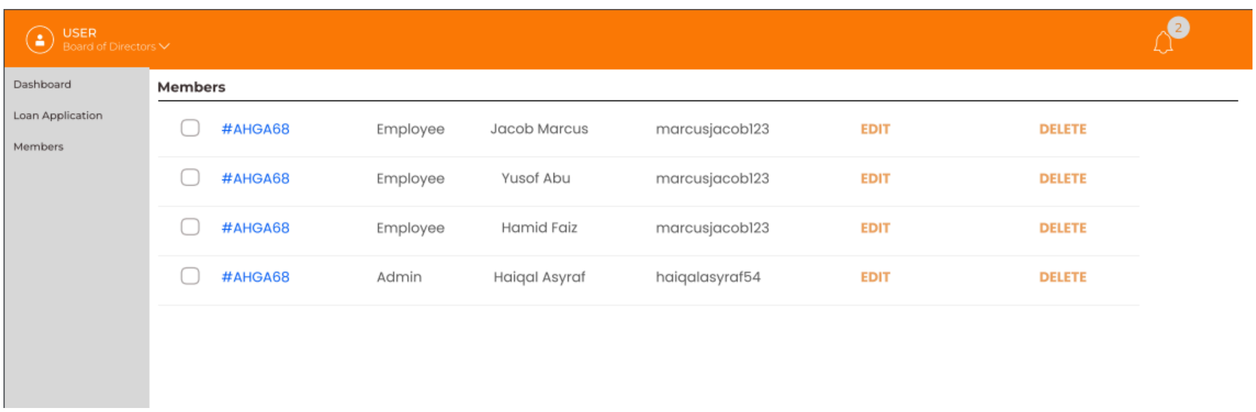
<u>ADMINISTRATOR INTERFACE</u>





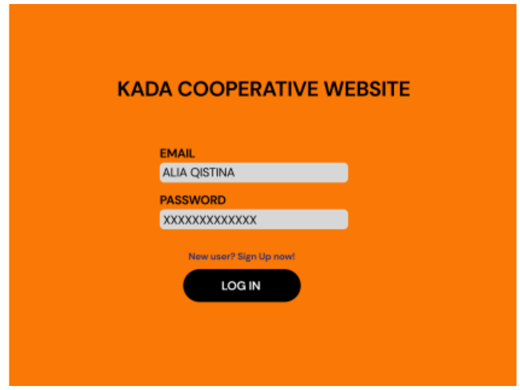

USER
Administrator
2
Dashboard
User Accounts
Loan Types
Loan Applicants
Member
User Accounts
User Details
Email
haziq999@gmail.com
User Type
Member
Name
Haziq Rusli Bin Hamdan
Done

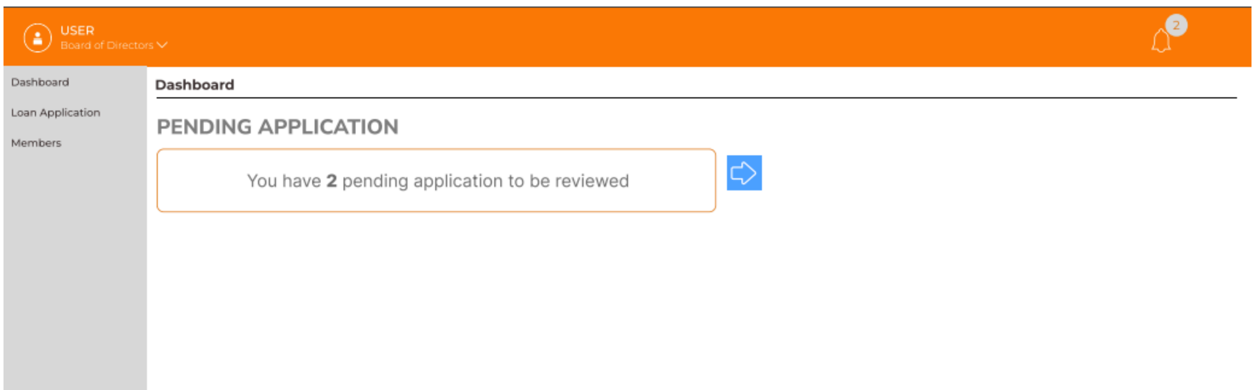

USER
Administrator
2
Dashboard
User Accounts
Loan Types
Loan Applicants
Member
Loan Types
Name
Repayment Type
Interest Rate
Maximum Loan
Al-Baiubithaman Ajil
Interest Only
5%
RM 15,000.00
REMOVE
UPDATE
Bai Al-Inah
Regular
3%
RM 10,000.00
REMOVE
UPDATE

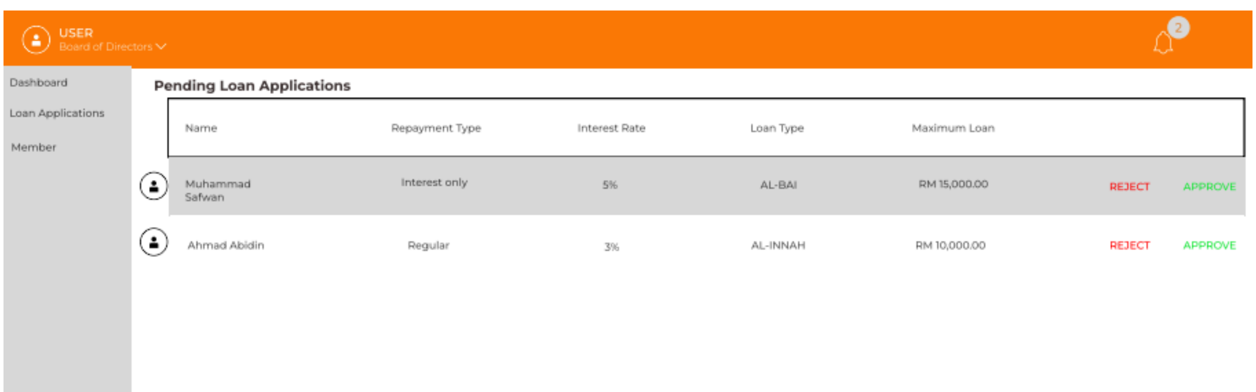

USER
Board of Directors
2
Dashboard
Loan Application
Members
Members
#AHGA68
Employee
Jacob Marcus
marcusjacob123
EDIT
DELETE
#AHGA68
Employee
Yusof Abu
marcusjacob123
EDIT
DELETE
#AHGA68
Employee
Hamid Faiz
marcusjacob123
EDIT
DELETE
#AHGA68
Admin
Haiqal Asyraf
haiqalasyraf54
EDIT
DELETE

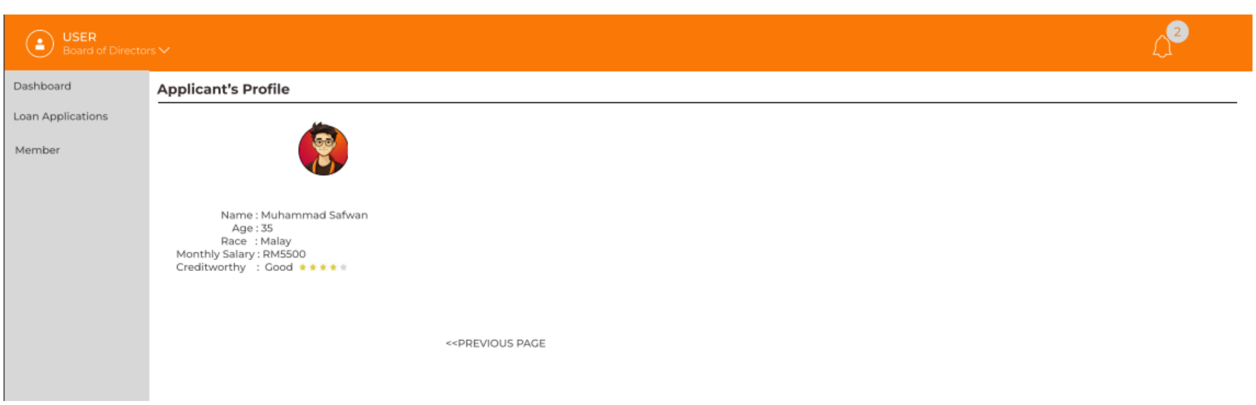
BOARD OF DIRECTORS INTERFACE

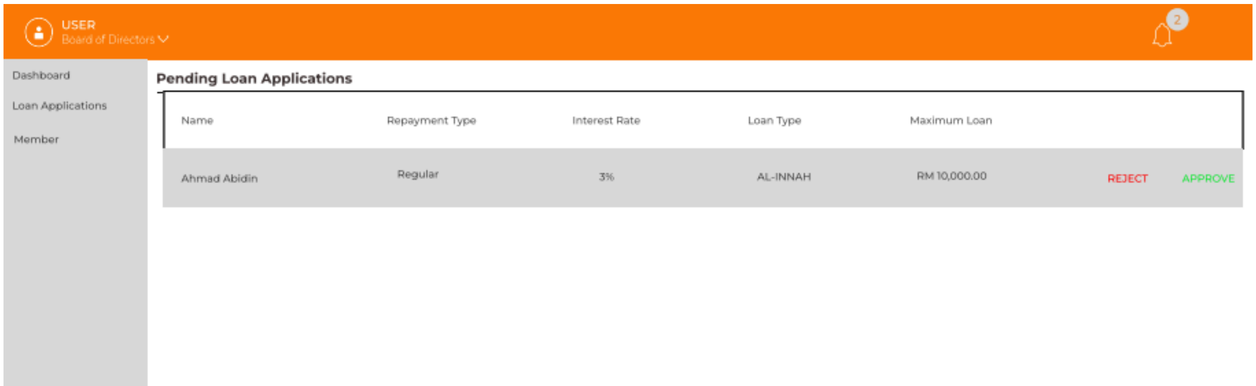
KADA
LEMBAGA KEMAJUAN PERTANIAN KEMUBU
KEMENTERIAN PERTANIAN DAN KETERJAMINAN MAKANAN

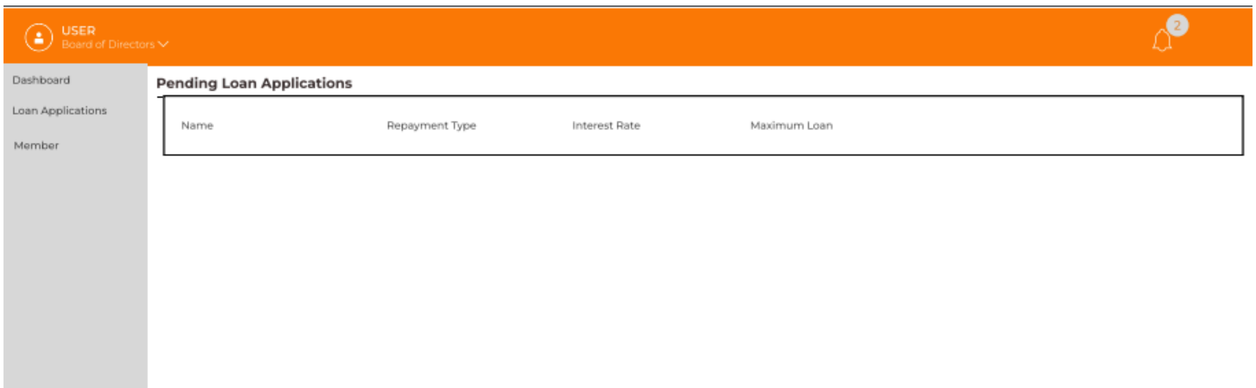


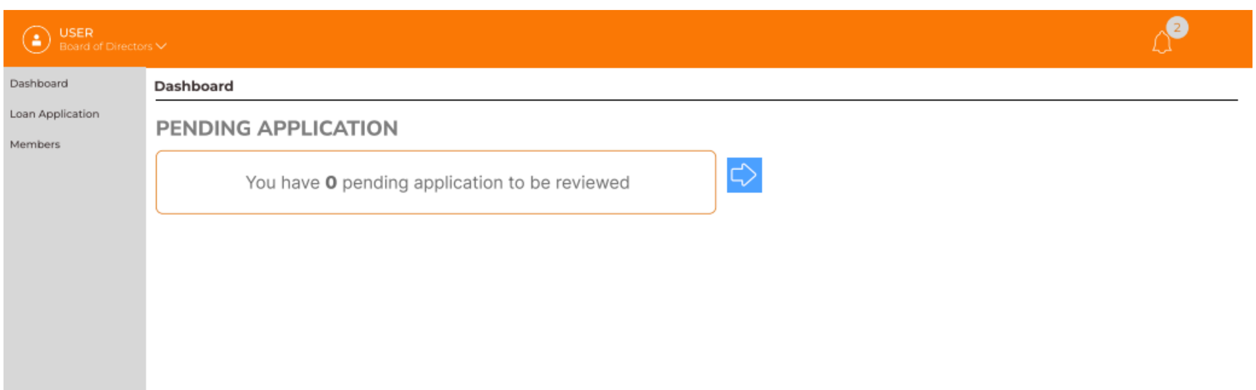


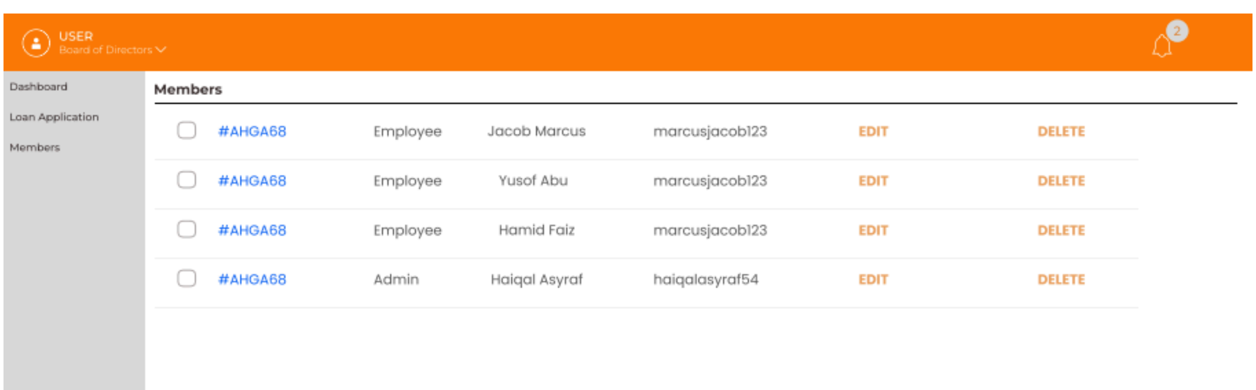





USER
Board of Directors
2
Dashboard
Loan Applications
Member
Pending Loan Applications
Name
Repayment Type
Interest Rate
Loan Type
Maximum Loan
Ahmad Abidin
Regular
3%
AL-INNAH
RM 10,000.00
REJECT
APPROVE


USER
Board of Directors
2
Dashboard
Loan Applications
Member
Pending Loan Applications
Name
Repayment Type
Interest Rate
Maximum Loan


USER
Board of Directors
2
Dashboard
Loan Application
Members
Dashboard
PENDING APPLICATION
You have 0 pending application to be reviewed


USER
Board of Directors
2
Dashboard
Loan Application
Members
Members
#AHGA68
Employee
Jacob Marcus
marcusjacob123
EDIT
DELETE
#AHGA68
Employee
Yusof Abu
marcusjacob123
EDIT
DELETE
#AHGA68
Employee
Hamid Faiz
marcusjacob123
EDIT
DELETE
#AHGA68
Admin
Haiqal Asyraf
haiqalasyraf54
EDIT
DELETE

4.5 SUMMARY

To summarise, this chapter successfully explains the analysis and design process of the project as phase two. Starting with the collection of system requirements to establish what the present system inclusive of its characteristics and process looks like, this will involve a company background. On using the digital flow diagrams, one can determine the business flow currently in practice to streamline the components of the new system. In the following sections, a new process is proposed, and a digital flow diagram (to-be) along with child diagrams shows it. Process specifications help elaborate the roles that are vague; partitioning makes it easier to deal with the data. A CRUD matrix defines users' activities and access rights, and an event response table considers the interactions. The structure chart depicts system elegance, while a system architectural diagram defines the technologies and data to be used. The system design is then turned into a non-functioning model by implementing input and output design methods for the interface.

REFERENCES

1. Lie, R. E. Y. N. A. L. D. I., & Fajar, A. N. (2022). Analysis And Development Of Microservices Architecture In Loan Application System Of Cooperative Enterprise In Indonesia. *Journal of Theoretical and Applied Information Technology*, *100*(23).http://www.jatit.org/volumes/Vol100No23/22Vol100No23.pdf

2. Noranita, B., Nugraheni, D. M. K., Fitriyani, M. I., & Nurhayati, Y. (2021, July). Business architecture and information system architecture design in savings and payment unit Koperasi Pegawai Republik Indonesia (KPRI) Diponegoro University using TOGAF 9 framework. In *Journal of Physics: Conference Series* (Vol. 1943, No. 1, p. 012105). IOP Publishing. https://iopscience.iop.org/article/10.1088/1742-6596/1943/1/012105/pdf

3. Zwass, V. (2001, October 12). *Information system | Definition, Examples, & Facts*. Encyclopedia Britannica. https://www.britannica.com/topic/information-system https://www.britannica.com/topic/information-system

4. CSM Technologies. (n.d.). *Cooperative Management Information System*. https://www.csm.tech/government-functions/offering/cooperative-management-information-system/

5. Suwandi, S., Asfi, M., Novelia, B., & Gitama, G. N. D. P. (2024, April). Digitalization of Employee Cooperative Loan Applications. In *Proceeding International Conference on Religion, Science and Education* (Vol. 3, pp. 943-952). https://sunankalijaga.org/prosiding/index.php/icrse/article/download/1279/1108

APPENDIX 1
MEMBERSHIP APPLICATION FORM

KOPERASI KAKITANGAN KADA KELANTAN BERHAD

PERMOHONAN MENJADI ANGGOTA

MAKLUMAT PEMOHON

NAMA

NO. KP

TARAF PERKAHWINAN

ALAMAT RUMAH

POSKOD

NEGERI

NO. ANGGOTA

NO. PF

JAWATAN & GRED

ALAMAT PEJABAT (Tempat Bertugas)

POSKOD

NO. TEL/FAX

NO. TEL BIMBIT

NO. TEL RUMAH

GAJI BULANAN RM .

JANTINA : LELAKI / PEREMPUAN

AGAMA : ISLAM / LAIN-LAIN

Nyatakan

BANGSA : MELAYU / LAIN-LAIN

Nyatakan

MAKLUMAT KELUARGA DAN PEWARIS

BIL	HUBUNGAN	NAMA	NO. K/P @ NO. SRT BERANAK

* SILA BUAT LAMPIRAN JIKA RUANGAN YANG DISEDIAKAN TIDAK MENCUKUPI

YURAN DAN SUMBANGAN

JIKA DI TERIMA SEBAGAI ANGGOTA, SAYA BERSETUJU MEMBAYAR YURAN DAN SUMBANGAN BULANAN SEPERTI DI BAWAH :

BIL	PERKARA	RM
1	FEE MASUK	
2	MODAH SYER *	
3	MODAL YURAN	
4	WANG DEPOSIT ANGGOTA	
5	SUMBANGAN TABUNG KEBAJIKAN (AL-ABRAR)	
6	SIMPANAN TETAP	
7	LAIN-LAIN	

* MINIMA MODAL SYER ADALAH SEBANYAK RM 300.00 DAN TIDAK MELEBIHI 1/5 DARIPADA MODAL SYER KOPERASI DAN HENDAKLAH DIJELASKAN DALAM TEMPOH 6 BULAN DARI TARIKH KELULUSAN MENJADI ANGGOTA.

APPENDIX 2
MEMBERSHIP FUNDING APPLICATION
PAGE 1

KOPERASI KAKITANGAN KADA KELANTAN BERHAD

Borang KKKK 02/00

PERMOHONAN PEMBIAYAAN ANGGOTA

BUTIR-BUTIR PEMBIAYAAN

JENIS PEMBIAYAAN
- ☐ AL - BAI
- ☐ AL-INAH
- ☐ SKIM KHAS
- ☐ KARNIVAL MUSIM ISTIMEWA
- ☐ BAIK PULIH KENDERAAN
- ☐ CUKAI JALAN
- ☐ LAIN-LAIN

Nyatakan ____________

AMAUN DIPOHON RM ☐☐☐☐☐ . ☐☐

TEMPOH PEMBIAYAAN ☐☐☐ BULAN

ANSURAN BULAN RM ☐☐☐ . ☐☐

BUTIR-BUTIR PERIBADI PEMOHON

NAMA

NO. KP

TARIKH LAHIR — UMUR — TAHUN

ALAMAT RUMAH

POSKOD — BANDAR

NEGERI

NO. ANGGOTA

NO. PF

JAWATAN

ALAMAT PEJABAT (Tempat Bertugas)

POSKOD — BANDAR

NO. TEL/FAX

NO. TEL BIMBIT

NAMA BANK/CWG

NO. AKAUN BANK

JANTINA ☐ LELAKI ☐ PEREMPUAN

AGAMA ☐ ISLAM ☐ LAIN-LAIN

Nyatakan ________

BANGSA ☐ MELAYU ☐ LAIN-LAIN

Nyatakan ________

PENGAKUAN PEMOHON

Saya ____________ No . K/P : ____________ dengan ini memberi kuasa kepada KOPERASI KAKITANGAN KADA KELANTAN BHD atau wakilnya yang sah untuk mendapat apa-apa maklumat yang diperlukan dan juga mendapatkan bayaran balik dari potongan gaji dan emolumen saya sebagaimana amaun yang dipinjamkan. Saya juga bersetuju menerima sebarang keputusan dari KOPERASI ini untuk menolak permohonan tanpa memberi sebarang alasan.

TANDATANGAN

Tarikh :

APPENDIX 2
MEMBERSHIP FUNDING APPLICATION
PAGE 2

BUTIR-BUTIR PENJAMIN

PENJAMIN 1

NAMA ____________________

NO. KP ____________ NO. PF ______

NO. ANGGOTA ____________________

PENJAMIN 2

NAMA ____________________

NO. KP ____________ NO. PF ______

NO. ANGGOTA ____________________

Adalah dengan ini saya/kami dengan rela hati bersetuju bersama-sama atau berasingan bertanggungjawab sepenuhnya ke atas ansuran harga barangan **atau pembiayaan ini. Kami** berjanji akan membayar **balik semua ansuran hutang** yang diberikan kepada peminjam di atas jika sekiranya beliau tidak dapat membayar balik ansuran itu mengikut perjanjian yang telah dipersetujui.

Tandatangan Penjamin 1
Bertarikh :

SAKSI 1
No.K/P :
No. Telefon :

Tandatangan Penjamin 2
Bertarikh :

SAKSI 2
No.K/P :
No. Telefon :

PENGESAHAN MAJIKAN

Kami mengesahkan bahawa : ____________________ No . K/P : ____________

telah memberikan butir-butir peribadi dan maklumat pendapatan selaras dengan rekod pekerjaan kakitangan tersebut.

Kami juga mengesahkan bahawa kakitangan adalah berjawatan tetap.

Gaji pokok sebulan RM

Gaji bersih sebulan RM

Tandatangan Sah/Cop

COP

Bertarikh :

KEGUNAAN PEJABAT

Tarikh Permohonan diterima / /

Jumlah Syer dan Yuran Terkumpul

Jumlah baki pinjaman sedia ada :

AL - BAI

AL-INAH

SKIM KHAS

CUKAI JALAN

BAIK PULIH KENDERAAN

KARNIVAL MUSIM ISTIMEWA

LAIN-LAIN (Nyatakan)

Bil. Permohonan

Jumlah Potongan Koperasi

Nota Akaun :

<u>Debit</u>		<u>Kredit</u>		<u>Amaun</u>
1 Belian	______	Bank	______	______
2 Si berhutang		Jualan		

Tandatangan Pengurus Besar
Bertarikh :

<u>Keputusan Mesyuarat Jawatan Kuasa Perniagaan</u>

DILULUSKAN

TIDAK LULUS

Bilangan Mesyuarat /

Tarikh / /

Catatan

Tandatangan
Pengerusi J/K Perniagaan
Bertarikh :

APPENDIX 3
FINANCIAL STATEMENTS OF COOPERATIVE MEMBERS

NAMA: MOHD ROSLAN B ISMAIL

NO. K/P: 740807-03-5479 **NO. PF:** 2273

NO. AHLI: 1579

Tuan/Puan,

PENGESAHAN PENYATA KEWANGAN AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023.

Untuk penentuan Juruaudit, kami dengan ini menyatakan bagi akaun tuan/puan adalah sebagaimana berikut:

MAKLUMAT SAHAM AHLI:

Modah Syer:	Modal Yuran:	Simpanan Tetap:	Tabung Anggota:	Simpanan Anggota:
RM	RM	RM	RM	RM

MAKLUMAT PINJAMAN AHLI:

Al-Bai:	Al-Innah:	B/Pulih Kenderaan:	Road Tax& Insuran:	Khas:	Al-Qadrul Hassan:
RM0.00	RM0.00	RM0.00	RM0.00	RM0.00	RM0.00

PENGESAHAN BAGI PENYATA KEWANGAN AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023

Saya ....MOHD ROSLAN B ISMAIL........ No. Ahli: ....1579........ mengesahkan bahawa Penyata Kewangan Koperasi Kakitangan KADA Kelantan Berhad bagi tahun berakhir 30 Jun 2023 adalah benar:

Tandatangan: ....

Tarikh: ....

☐ Setuju

☐ Tidak Setuju..............................
(Nyatakan sebabnya)

Nota:
1. Sila tandakan (/) di salah satu kotak diatas.
2. Sekiranya pihak Koperasi tidak menerima sebarang maklumbalas daripada tuan/puan sehingga 30 Jun 2023, maka pengesahan penyata ini dianggap betul dan tuan/puan bersetuju.

APPENDIX 4
LOAN REPAYMENT CHART

JADUAL PEMBAYARAN BALIK PEMBIAYAAN SKIM AL-BAI'UBITHAMAN AAJIL/BAI AL-INAH

KADAR KEUNTUNGAN	4.20%					
TEMPOH (TAHUN)	1	2	3	4	5	6
TEMPOH (BULANAN)	12	24	36	48	60	72
JUMLAH PEMBIAYAAN	ANSURAN BULANAN					
1,000.00	86.83	45.17	31.28	24.33	20.17	17.39
2,000.00	173.67	90.33	62.56	48.67	40.33	34.78
3,000.00	260.50	135.50	93.83	73.00	60.50	52.17
4,000.00	347.33	180.67	125.11	97.33	80.67	69.56
5,000.00	434.17	225.83	156.39	121.67	100.83	86.94
6,000.00	521.00	271.00	187.67	146.00	121.00	104.33
7,000.00	607.83	316.17	218.94	170.33	141.17	121.72
8,000.00	694.67	361.33	250.22	194.67	161.33	139.11
9,000.00	781.50	406.50	281.50	219.00	181.50	156.50
10,000.00	868.33	451.67	312.78	243.33	201.67	173.89
11,000.00	955.17	496.83	344.06	267.67	221.83	191.28
12,000.00	1042.00	542.00	375.33	292.00	242.00	208.67
13,000.00	1128.83	587.17	406.61	316.33	262.17	226.06
14,000.00	1215.67	632.33	437.89	340.67	282.33	243.44
15,000.00	1302.50	677.50	469.17	365.00	302.50	260.83
16,000.00	1389.33	722.67	500.44	389.33	322.67	278.22
17,000.00	1476.17	767.83	531.72	413.67	342.83	295.61
18,000.00	1563.00	813.00	563.00	438.00	363.00	313.00
19,000.00	1649.83	858.17	594.28	462.33	383.17	275.06
20,000.00	1736.67	903.33	625.56	486.67	403.33	347.78

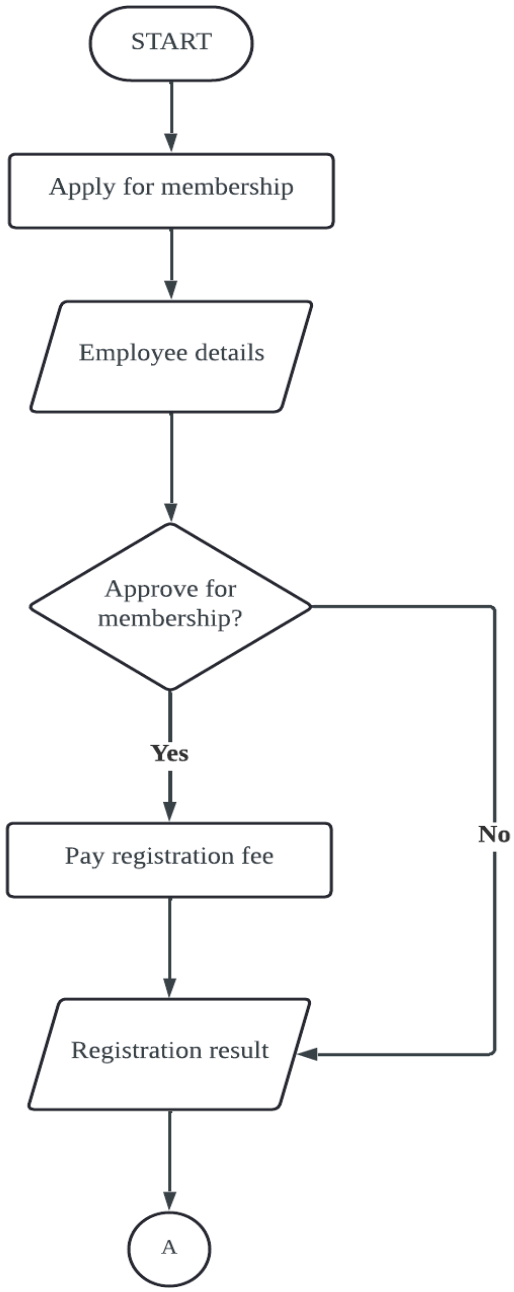
Dikemaskini 10 Mei 2023

Mesyuarat Lembaga bil 3/2023 7 Mei 2023

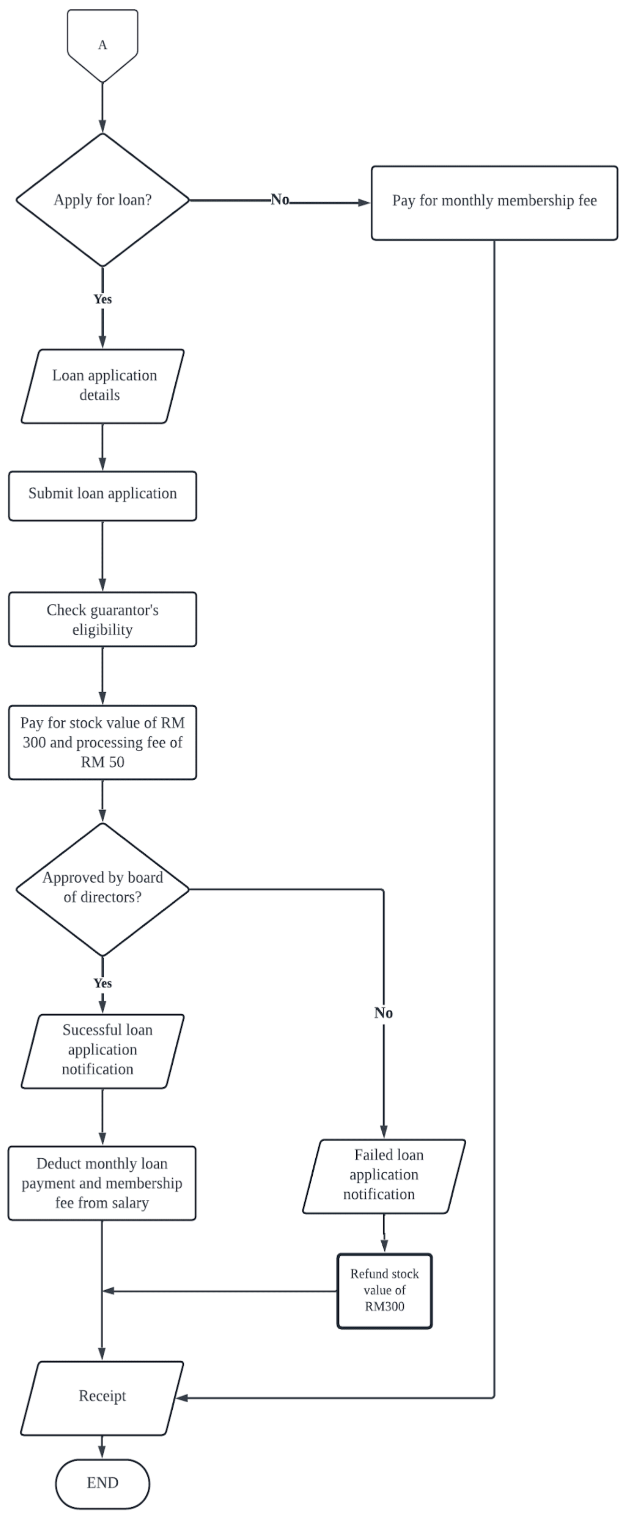
APPENDIX 5
GANTT CHART

Contribution	Activity	Duration (days)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
Afif, Afiq, Alif, Iman	Collecting information from interview	3																																
Afif, Afiq, Alif, Iman	Meeting with team member to plan and divide task	1																																
Afif, Afiq, Alif, Iman	Extracting key information from the interview	1																																
Afif, Afiq, Alif, Iman	Determine user requirement for the proposed system	2																																
Afif, Afiq, Alif, Iman	Planning a schedule of the project	1																																
Afif, Afiq, Alif, Iman	Choose the Methodology	1																																
Iman	Determine objective of the study	1																																
Iman	Comparing the previous research	1																																
Afif, Afiq, Alif, Iman	Create a Project Planning Schedule	5																																
Afiq, Iman	Determine problem statement based on interview	2																																
Iman	Proposed solutions based on problem statement	2																																
Afif, Afiq, Alif, Iman	Describe the current business process	1																																
Iman	Determine functional and non-functional requirements of the system	1																																
Afif, Afiq, Alif, Iman	Design a Logical DFD of AS-IS system	3																																
Afif, Afiq, Alif, Iman	Design a Logical DFD (TO BE) system	3																																
Afif, Afiq, Alif, Iman	Design a Physical DFD (TO BE) system	3																																
Afiq	Child diagram for process 1.0																																	
Afiq	Child diagram for process 2.0																																	
Alif	Child diagram for process 3.0																																	
Afif	Child diagram for process 4.0																																	
Afif	Child diagram for process 5.0																																	
Iman	Child diagram for process 6.0																																	
Alif	Child diagram for process 7.0																																	
Afif, Afiq, Alif, Iman	Design a Partitioning	1																																
Afif, Alif, Iman	Create a CRUD Matrix and System Architecture	1																																
Iman	Create an Event Response Table, and Structure Chart	2																																
Afif,Alif	Make a Interface Design / Prototype	5																																
Iman	Summary the used method	2																																

APPENDIX 6
FLOWCHART OF CURRENT BUSINESS PROCESS

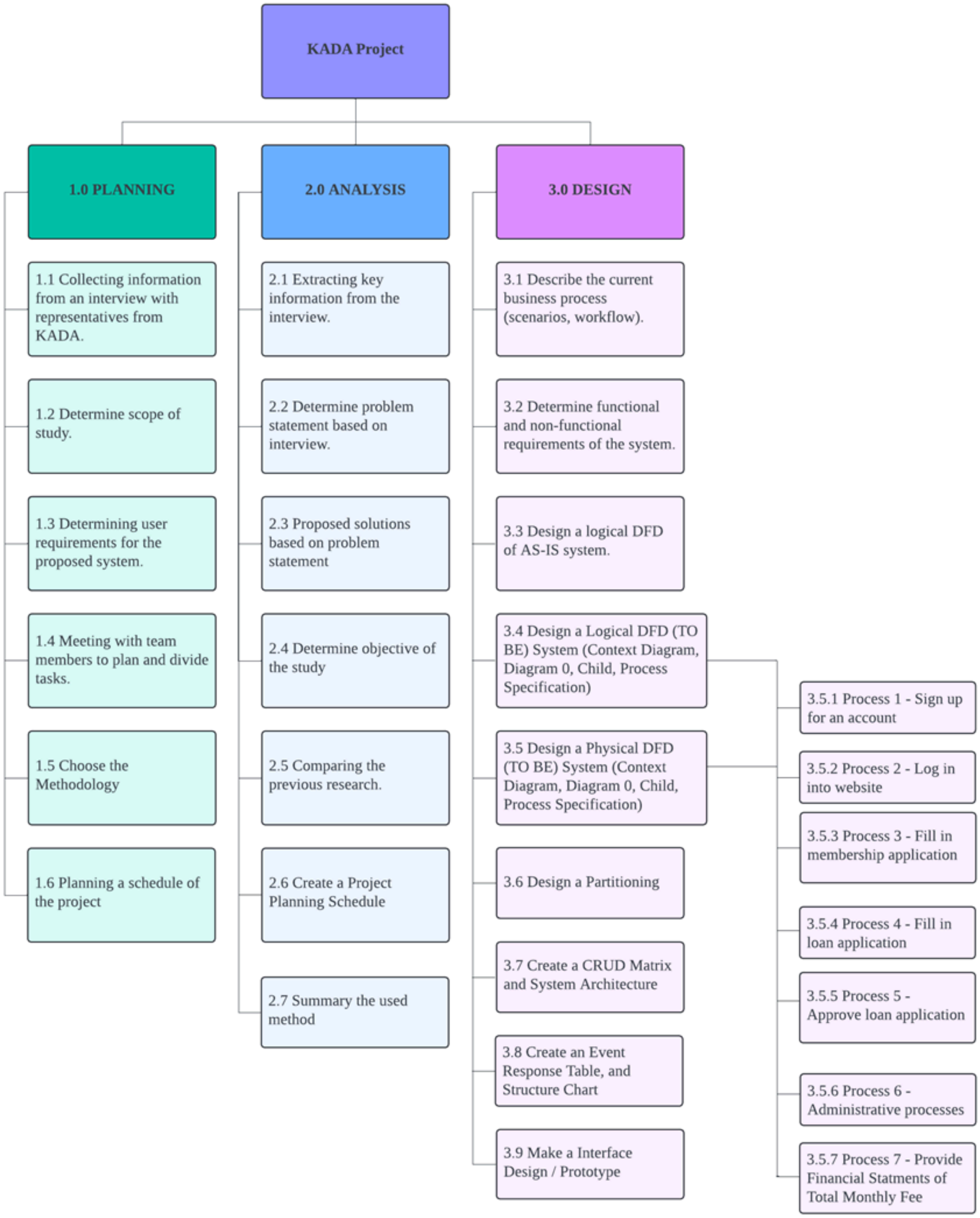


APPENDIX 7
FLOWCHART OF CURRENT BUSINESS PROCESS



APPENDIX 8

WORK-BASED STRUCTURE DIAGRAM



APPENDIX 9
PERT DIAGRAM